Main Routine

EDC2 - v2.01 add Event Configuration for DtOwner and DtType
v3.50 - L980 Controller; Summary KPI additions for ProdDelay, ToolChange and Configurable Option 1-5

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	101
Dest	Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1103711128
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1103711128

Configuration

2

S:FS

EDC Config Configuration ONS (Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_111_Config

Main Routine

Search ScheduleLookkup for the first HourStart where SysTime (HHMMSS) > HourStart (HHMMSS)
Store the Schedule, SchedulePrev and detect NewHour and ShiftReset

this only needs to execute on each new second since schedule can only change on the second (and typically only changes on the minute)
at end of shift, send final packet and then reset Summary data

3

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Source B ScheduleTimePrev
151833

Schedule Lookup
Time of Prev Check

MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest ScheduleTimePrev
151833

Schedule Search FSC
ScheduleFSC
RES

Schedule Search FSC

MOVE
Source 0
Dest ScheduleFSC.POS
16

Schedule Search FSC

ADD
Source A CfgSched.ei
22
Source B 1
Dest ScheduleFSC.LEN
23

Schedule Search FSC

FSC
Control ScheduleFSC
Length 23
Position 16
Mode ALL
Expression CLX.SysTime[9]<ScheduleLookup[ScheduleFSC.POS].HourStart
EN
DN
ER

Schedule Search FSC
ScheduleFSC.FD

Schedule Lookup
Array Position

SUB
Source A ScheduleFSC.POS
16
Source B 1
Dest SchedulePOS
15

Schedule
Previous Second

COP
Source Schedule
Dest SchedulePrev
Length 1

Schedule
Current Hour

COP
Source ScheduleLookup[SchedulePOS]
Dest Schedule
Length 1

Schedule
Current Hour Hour
(0-10)

NE
Source A Schedule.Hour
9
Source B SchedulePrev.Hour
9

Schedule
New Hour
ScheduleNewHour

diagnostic

ADD
Source A SchedNewHourCount
3129
Source B 1
Dest SchedNewHourCount
3129

Schedule
Current Hour Shift
Reset (0,1)

EQ
Source A Schedule.Reset
0
Source B 1

Schedule
Current Hour Shift
Reset (0,1)

NE
Source A Schedule.Reset
0
Source B SchedulePrev.Reset
0

Schedule
Shift Reset
ScheduleShiftReset
L

diagnostic

ADD
Source A SchedShiftResetCount
285
Source B 1
Dest SchedShiftResetCount
285

Main Routine

4

Load Summary Ring Buffer Timer
periodically send EQ, Line and Group Summary if the Ring Buffer is empty
(no point in adding more Summary data if processing is backlogged - it will be old data when processed)
NewHour must be logged to close out the existing hour (bypass the buffer empty check)

Send the data on the minute change for a consistent interval

Load Summary Ring Buffer Timer
MOVE
Source 65000
Dest SumDataTMR.PRE
65000

Load Summary Ring Buffer Timer
TON
Timer SumDataTMR
Preset 65000
Accum 53558
EN
DN

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)
NE
Source A CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Source B SysTimePrevMinute
24

MOVE
Source CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Dest SysTimePrevMinute
24

ScheduleNewMinute
L

Load Summary Ring Buffer Timer
SumDataTMR.DN

ScheduleNewMinute

Schedule New Hour
ScheduleNewHour

EDC Ring Buffer Control Buffer Empty
RingBufferSum01.Empty
<EDC.RB_Sum01.Empty>

EDC Ring Buffer Control Buffer Empty
RingBufferSum02.Empty
<EDC.RB_Sum02.Empty>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RingBufferSum01.NotEmptyTMR.EN
<EDC.RB_Sum01.NotEmptyTMR.EN>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RingBufferSum02.NotEmptyTMR.EN
<EDC.RB_Sum02.NotEmptyTMR.EN>

Load Summary Ring Buffer by LNi
FLL
Source -1
Dest LineSumDataLoad[0]
Length 4

Load Summary Ring Buffer Timer
SumDataTMR
RES

ScheduleNewMinute
U

Main Routine

Use Schedule Shift Reset signal to reset all shift-based data
delay Reset by 1 scan to allow NewHour to send end-of-shift data before resetting values
(Reset and NewHour are enabled on same scan)

5

Schedule Shift Reset
ScheduleShiftReset

Schedule New Hour
ScheduleNewHour

Schedule Shift Reset
ScheduleShiftReset
(U)

Reset Line, Group and Equip Data by LNi

FLL
Source -1
Dest LineDataReset[0]
Length 4

ShiftResetONS
(ONS)

ShiftReset

Line Loop Configuration Index

When LNi = CfgLNi, all Equipment are checked for Callout Messages (to find new Equipment)
When LNi <> CfgLNi, only configured Equipment are checked for Callout Messages
Only 1 Line/Scan is checked for new Equipment to save scan time
(only increment if crrent CfgLNi was in Lines scanned during last Line Loop)

6

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Config Update Line Index

LIMIT
Low Limit LineLoopInit 15
Test CfgLNi 8
High Limit LineLoopTerm 14

Config Update Line Index

ADD
Source A CfgLNi 8
Source B 1
Dest CfgLNi 8

Config Update Line Index

LT
Source A CfgLNi 8
Source B 0

Config Update Line Index

GT
Source A CfgLNi 8
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Config Update Line Index

MOVE
Source 0
Dest CfgLNi 8

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Config Update Line Index

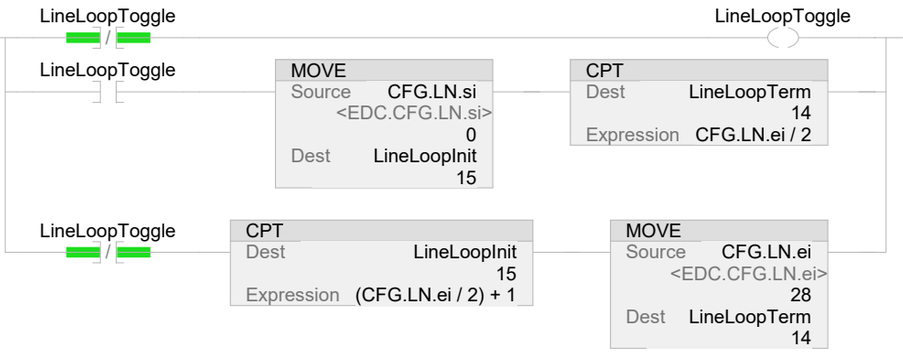
MOVE
Source _ViewLNi 9
Dest CfgLNi 8

Main Routine

Line Loop Initial/Terminal values for n Lines/Scan
reduce scan time by updating portion of lines each scan but maintain <200 msec Data Increment

Loop Increment = 0-20, 21-40
2 scans update all Lines

7

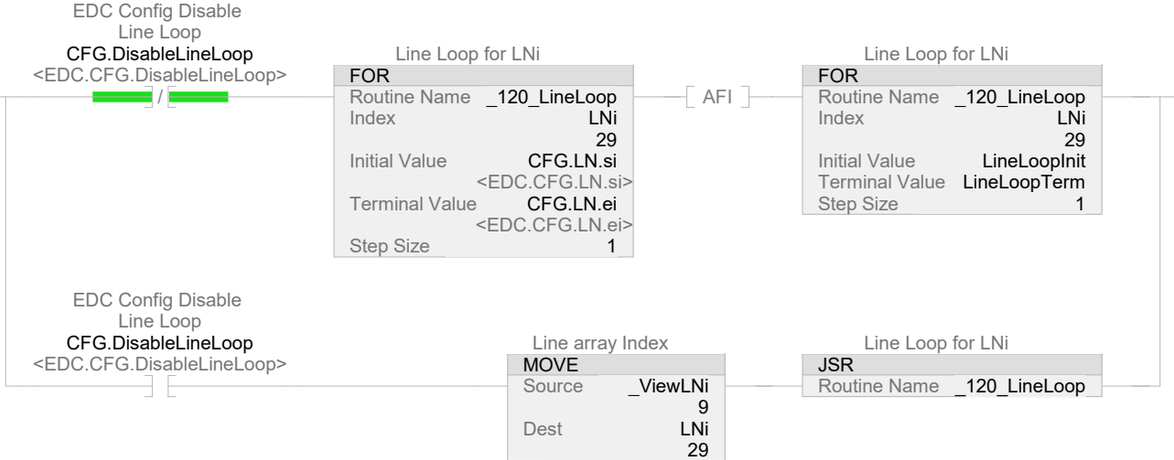


Line Loop
service all Lines each Scan for fast update of Live Callout table

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

DHF008 -- service all Lines every scan (was using 2 scans)

8



Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

9

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	152
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	4
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	20
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	19
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9003
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

10

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

11

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
3696
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.1
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

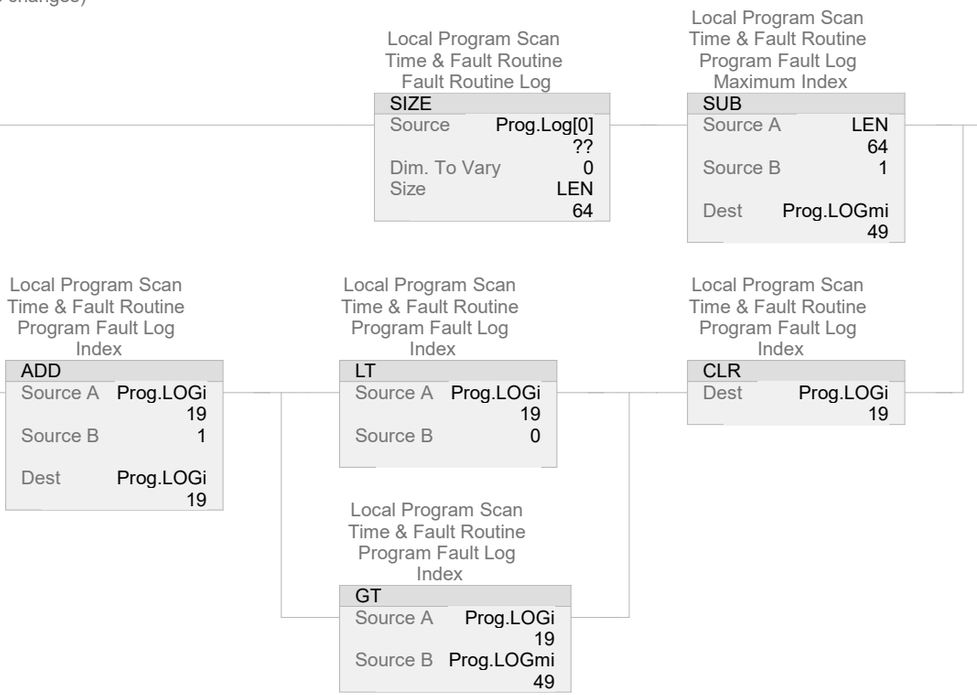
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
3

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
143
Dest Prog.Log[Prog.LOGi].Routine
152

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
135064652

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20220707

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
110119

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
659

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source LNi
29
Dest Prog.Log[Prog.LOGi].Opt1
28

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source EQi
12
Dest Prog.Log[Prog.LOGi].Opt2
82

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 102
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 588226492

(End)

Configuration

Callout Message Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 111
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Configuration

Callout Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

2

Equipment Data Buffer Callout/Status Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Callout[0]
1
Dim. To Vary 0
Size CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14

EDC Config Array Max Index

SUB
Source A CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14
Source B 1
Dest CFG.CalloutMSG.mi
<EDC.CFG.CalloutMSG.mi>
13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14

EDC Config Array End Count (udt size)

GT
Source A CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14
Source B CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14

EDC Config Array End Count (udt size)

MOVE
Source CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14
Dest CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14
Source B 1
Dest CFG.CalloutMSG.ei
<EDC.CFG.CalloutMSG.ei>
13

Configuration

Callout Table Array
Configure Array Indexes

3

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

SIZE
Source EDC_EqCallout[0,0] ??
Dim. To Vary 0
Size CFG.CalloutLN.n <EDC.CFG.CalloutLN.n> 29

EDC Config Array Max Index

SUB
Source A CFG.CalloutLN.n <EDC.CFG.CalloutLN.n> 29
Source B 1
Dest CFG.CalloutLN.mi <EDC.CFG.CalloutLN.mi> 28

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

SIZE
Source EDC_EqCallout[0,0] ??
Dim. To Vary 1
Size CFG.CalloutEQ.n <EDC.CFG.CalloutEQ.n> 100

EDC Config Array Max Index

SUB
Source A CFG.CalloutEQ.n <EDC.CFG.CalloutEQ.n> 100
Source B 1
Dest CFG.CalloutEQ.mi <EDC.CFG.CalloutEQ.mi> 99

Callout Table Array Size
Predefined DINT + 40 Equipment-Defined DINTs (8 DINTs * 5 Packets)

4

Local Copy of EDC_EqCallout for this EQID

SIZE
Source TempCallout[0] 16#0000_00a3
Dim. To Vary 0
Size CFG.CalloutARR.n <EDC.CFG.CalloutARR.n> 65

EDC Config Array Max Index

SUB
Source A CFG.CalloutARR.n <EDC.CFG.CalloutARR.n> 65
Source B 1
Dest CFG.CalloutARR.mi <EDC.CFG.CalloutARR.mi> 64

Configuration

Callout Packet Count
v1.3 Fault/Callout MultiPacket support

EDC Config Array Size

MOVE
Source 8
Dest CFG.CalloutPKT.n <EDC.CFG.CalloutPKT.n> 8

EDC Config Array Max Index

SUB
Source A CFG.CalloutPKT.n <EDC.CFG.CalloutPKT.n> 8
Source B 1
Dest CFG.CalloutPKT.mi <EDC.CFG.CalloutPKT.mi> 7

Callout Packet Eq-Defined Array Size (8 DINTs/Packet)
v1.3 Fault/Callout MultiPacket support

Equipment Data Callout/Status Message Equipment Defined Status Words

SIZE
Source CalloutMsg.EqDef.Word[0] 16#0000_0000
Dim. To Vary 0
Size CFG.CalloutEqDef.n <EDC.CFG.CalloutEqDef.n> 8

EDC Config Array Max Index

SUB
Source A CFG.CalloutEqDef.n <EDC.CFG.CalloutEqDef.n> 8
Source B 1
Dest CFG.CalloutEqDef.mi <EDC.CFG.CalloutEqDef.mi> 7

Zero Array used for FBC Comparison

SIZE
Source ZeroArray[0] 0
Dim. To Vary 0
Size LEN 64

FLL
Source 0
Dest ZeroArray[0]
Length LEN

Initialize Line View

_ViewEN
U

Configuration

Schedule Array

9

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>
U

Schedule Hours lookup array Shift Configuration from EDC Database

SIZE
Source ScheduleLookup[0] ??
Dim. To Vary 0
Size CfgSched.n 100

Schedule Array Config Array Max Index

SUB
Source A CfgSched.n 100
Source B 1
Dest CfgSched.mi 99

Schedule Array Config Array Start Index

MOVE
Source 0
Dest CfgSched.si 0

Schedule Array Config Array End Index

MOVE
Source CfgSched.mi 99
Dest CfgSched.ei 22

schedule downloaded from server
no longer need hard-coded schedule

10

lbl_Exit
JMP

Configuration

hardcode Schedule Shift Configuration
(this will be replaced by a configuration download from EDC database)

11

Schedule Hours lookup array Shift Configuration from EDC Database

FLL
Source -1
Dest ScheduleLookup[0]
Length CfgSched.n

Schedule Array Config Array Start Index

MOVE
Source 0
Dest CfgSched.si 0

Schedule Array Config Array End Index

MOVE
Source 22
Dest CfgSched.ei 22

12

MOVE
Source 0
Dest index 22

Schedule Hours lookup array

MOVE
Source 0
Dest ScheduleLookup[index].HourStart 240000

Schedule Hours lookup array

MOVE
Source 2
Dest ScheduleLookup[index].Shift 2

Schedule Hours lookup array

MOVE
Source 3
Dest ScheduleLookup[index].Qtr 2

Schedule Hours lookup array

MOVE
Source 5
Dest ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE
Source 0
Dest ScheduleLookup[index].Reset 0

Configuration

13

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	1	Source	10000	Source	2	Source	3	Source	6
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

14

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	2	Source	21200	Source	2	Source	4	Source	7
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

15

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	3	Source	31200	Source	2	Source	4	Source	8
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

16

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	4	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	40500	Source	2	Source	5	Source	9
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

17

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	5	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	50500	Source	2	Source	5	Source	10
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

18

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	6	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	60000	Source	1	Source	0	Source	0
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	1
Dest	ScheduleLookup[index].Reset 0

Configuration

19

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	7	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	63600	Source	1	Source	1	Source	1
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

20

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	8	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	73600	Source	1	Source	1	Source	2
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

21

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	9	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	84200	Source	1	Source	2	Source	3
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

22

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	10		Source	94200		Source	1		Source	2		Source	4
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array	MOVE	
	Source	0
	Dest	ScheduleLookup[index].Reset 0

23

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	11		Source	111000		Source	1		Source	3		Source	5
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array	MOVE	
	Source	0
	Dest	ScheduleLookup[index].Reset 0

24

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	12		Source	121000		Source	1		Source	3		Source	6
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array	MOVE	
	Source	0
	Dest	ScheduleLookup[index].Reset 0

Configuration

25

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	13	Source	132700	Source	1	Source	4	Source	7
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

26

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	14	Source	142700	Source	1	Source	4	Source	8
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

27

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	15	Source	152500	Source	1	Source	5	Source	9
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

28

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	16	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	162500	Source	1	Source	5	Source	10
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

29

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	17	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	170000	Source	2	Source	0	Source	0
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	1
Dest	ScheduleLookup[index].Reset 0

30

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	18	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	192100	Source	2	Source	1	Source	1
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

31

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	19		Source	202100		Source	2		Source	1		Source	2
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

32

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	20		Source	212700		Source	2		Source	2		Source	3
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

33

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	21		Source	222700		Source	2		Source	2		Source	4
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

34

MOVE
Source 22
Dest index
22

Schedule Hours lookup array
MOVE
Source 240000
Dest ScheduleLookup[index].HourStart
240000

Schedule Hours lookup array
MOVE
Source 2
Dest ScheduleLookup[index].Shift
2

Schedule Hours lookup array
MOVE
Source 2
Dest ScheduleLookup[index].Qtr
2

Schedule Hours lookup array
MOVE
Source 4
Dest ScheduleLookup[index].Hour
4

Schedule Hours lookup array
MOVE
Source 0
Dest ScheduleLookup[index].Reset
0

35

lbl_Exit
LBL

NOP

(End)

Line Loop for LNi

Line Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 120
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Line Loop for LNi

Copy Line Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

Reset Line bits for Accumulated EQ status
Offline Line w/ multiple Data Groups (special case)
DN006 - support multiple Offline Lines

LineDataSTS - accumulates overlapping Status of Downtime types (Tool Change, Options, Full, Short, ProdDelay)
LineDataCTL - contains exclusive Control of Downtime Type (generated after all Group-Equipment processed)
STS & CTL are needed to handle seal-in of existing Downtime Type

2

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].CFG
Dest LineCfg
Length 1

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].CTL
Dest LineCtl
Length 1

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].Data
Dest LineData
Length 1

Local Copy of Line Array Tag for LNi
FLL
Source 0
Dest LineDataSTS
Length 1

Local Copy of Line Array Tag for LNi
FLL
Source 0
Dest LineDataCTL
Length 1

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].DataCTLprev
Dest LineDataCTLprev
Length 1

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Line array Index
EQ
Source A LNi
29
Source B CFG.OfflineFirstLine
<EDC.CFG.OfflineFirstLine>
0

Offline Group Data Initialize
FOR
Routine Name _141_OfflineGroupDataInit
Index GRPi
4
Initial Value CFG.GRP.si
<EDC.CFG.GRP.si>
Terminal Value CFG.GRP.ei
<EDC.CFG.GRP.ei>
Step Size 1

Initialize Offline Groups (1st Offline Line)
OfflineFirstLine

Line array Index
EQ
Source A LNi
29
Source B CFG.OfflineLastLine
<EDC.CFG.OfflineLastLine>
28

Finalize Offline Groups (Last Offline Line)
OfflineLastLine

Nov 24, 2025 - Preserve the CycleSend signal for sending Summary Data updates when still cycling after WorkTime is off

3

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].DataCTL.CycleSend

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend
L

Line Loop for LNi

v3.5 Preserve the Downtime Type seal-in

4

Equipment Data Collection Lines and Equipment Array	Local Copy of Line Array Tag for LNi
EDC.Line[LNi].DataCTL.ToolChange	LineDataCTL.ToolChange
EDC.Line[LNi].DataCTL.Option01	LineDataCTL.Option01
EDC.Line[LNi].DataCTL.Option02	LineDataCTL.Option02
EDC.Line[LNi].DataCTL.Option03	LineDataCTL.Option03
EDC.Line[LNi].DataCTL.Option04	LineDataCTL.Option04
EDC.Line[LNi].DataCTL.Option05	LineDataCTL.Option05
EDC.Line[LNi].DataCTL.ProdDelay	LineDataCTL.ProdDelay
EDC.Line[LNi].DataCTL.Full	LineDataCTL.Full
EDC.Line[LNi].DataCTL.Short	LineDataCTL.Short

Line Loop for LNi

Load & Reset Summary Ring Buffer Data signals for this Line

5

Line Bit Index

MOD
Source A: LNi 29
Source B: 32
Dest: LnBTi 28

Line Word Index

CPT
Dest: LnWDi 0
Expression: (LNi - LnBTi) / 32

Load Summary Ring Buffer by LNi: LineSumDataLoad[LnWDi].[LnBTi]

Load Summary Ring Buffer for this Line: SumDataLoad

Reset Line, Group and Equip Data by LNi: LineDataReset[LnWDi].[LnBTi]

Reset Summary Data for this Line: SumDataReset

Calculate Data Increment for this Line's Scan(s) since last update.
Previous Scan(s) Time increment will be added to all Line, Group and Equipment Values that were ON.
Each Line has its own DataInc to support Line updates over multiple scans (but for performance, Callout should run all Lines on a single scan if possible)

Wallclock CurrentValue microseconds is divided by 1000 for the millisecond data increment.
The millisecond data increment is multiplied by 1000 for the microseconds to add to the Previous Value (this <1000 microseconds accumulate for a future data increment)

6

Local Copy of Line Array Tag for LNi Data Increment (microseconds)

SUB
Source A: CLX.WallclockCurrentValue <EDC.CLX.WallclockCurrentValue> 1793391561904697
Source B: LineCtl.WallclockPreviousValue 6595304822000
Dest: LineCtl.DataIncMicrosec 16234

Local Copy of Line Array Tag for LNi Data Increment (milliseconds)

DIV
Source A: LineCtl.DataIncMicrosec 16234
Source B: 1000
Dest: LineCtl.DataInc 16

Local Copy of Line Array Tag for LNi Previous Value Increment (microseconds)

MUL
Source A: LineCtl.DataInc 16
Source B: 1000
Dest: LineCtl.PrevIncMicrosec 16000

Local Copy of Line Array Tag for LNi Wallclock Previous Value (microseconds)

ADD
Source A: LineCtl.WallclockPreviousValue 6595304822000
Source B: LineCtl.PrevIncMicrosec 16000
Dest: LineCtl.WallclockPreviousValue 6595304822000

Line Loop for LNi

Equipment Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

When LNi = CfgLNi, all Equipment are checked for Callout Messages (to find new Equipment)
When LNi <> CfgLNi, only configured Equipment are checked for Callout Messages
Only 1 Line/Scan is checked for new Equipment to save scan time

7

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>
/

Line array Index
NE
Source A: LNi 29
Source B: CfgLNi 8

Equipment Loop for EQi
FOR
Routine Name: _130_EquipLoop
Index: EQi 12
Initial Value: LineCfg.EQ.si
Terminal Value: LineCfg.EQ.ei
Step Size: 1

Line array Index
EQ
Source A: LNi 29
Source B: CfgLNi 8

Config Check Line LNi = CfgLNi
CfgLine

Temp Storage of LineCfg.EQ.ENei
CLR
Dest: LineCfg_EqENei_Temp 67

Equipment Loop for EQi
FOR
Routine Name: _130_EquipLoop
Index: EQi 12
Initial Value: 0
Terminal Value: CFG.EQ.mi <EDC.CFG.EQ.mi>
Step Size: 1

Local Copy of Line Array Tag for LNi Array End Index for Enabled Equipment
MOVE
Source: LineCfg_EqENei_Temp 67
Dest: LineCfg.EQ.ENei 99

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>

Equipment array Index
MOVE
Source: _ViewEQi 18
Dest: EQi 12

Equipment Loop for EQi
JSR
Routine Name: _130_EquipLoop

Faster to JMP over AOI if not scanned
(AOI uses scan time even when Rung is scanned false)

8

Reset Summary Data for this Line
SumDataReset
/

lbl_SkipReset
JMP

Line Loop for LNi

Shift Reset for Line & Offline OA
Fault & Alarm Time are not reset at Shift Change

Offline Line w/ multiple Data Groups (special case)
DN006 - support multiple Offline Lines

9

Reset Summary Data for this Line
SumDataReset

Data Reset for Line, Group, Equip

aoiEDC_DataReset	
aoiEDC_DataReset	aoiDataReset ...
Data	LineData
DataCtlPrev	LineDataCTLprev

Initialize Offline Groups (1st Offline Line)
OfflineFirstLine

Offline Group Data Reset

FOR	
Routine Name	_142_OfflineGroupDataReset
Index	GRPi
	4
Initial Value	CFG.GRP.si
	<EDC.CFG.GRP.si>
Terminal Value	CFG.GRP.ei
	<EDC.CFG.GRP.ei>
Step Size	1

Line Loop for LNi

Data Increment for Line and Offline OA
Add Incremental Time to Worktime, Downtime that are ON
Add Incremental Time to to Current Alarm or Fault Duration
Calculate OA
DN006 - support multiple Offline Lines

Oct 12, 2021 - don't update Offline until all Offline Lines are updated and data is complete
(work bit was toggling between Line 0 and Line 28 update and generating many WorkEnd signals)

10

lbl_SkipReset
LBL

Prioritize STS for CTL

aoiEDC_DataCtl

aoiEDC_DataCtl	aoiDataCtl
DataSts	LineDataSTS
DataCtl	LineDataCTL

Increment Data

aoiEDC_DataInc

aoiEDC_DataInc	aoiDataInc
Data	LineData
DataCtl	LineDataCTL
DataCtlPrev	LineDataCTLprev
Increment	LineCtl.DataInc

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Finalize Offline Groups (Last Offlline Line)
OfflineLastLine

Offline Group Data Update

FOR

Routine Name	_143_OfflineGroupData
Index	GRPi
	4
Initial Value	CFG.GRP.si
	<EDC.CFG.GRP.si>
Terminal Value	CFG.GRP.ei
	<EDC.CFG.GRP.ei>
Step Size	1

Line Loop for LNi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OA% - InDemand Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

11

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend

Schedule New Hour
ScheduleNewHour

Local Copy of Line Array Tag for LNi Work Time Ended
LineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 2
Dest SumMsg.Header.DataType 13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source LNi 29
Dest SumMsg.Header.ID 3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source LineData.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source LineData.EdtOA.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source LineData.EdtOA.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source LineData.LdtOA.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source LineData.LdtOA.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP
Source LineData.Full
Dest SumMsg.Data.Full

Equipment Data Summary Message 7 - Short

COP
Source LineData.Short
Dest SumMsg.Data.Short

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP
Source LineData.ProdDelay
Dest SumMsg.Data.ProdDelay

Equipment Data Summary Message 10 - Tool Change (ADC)

COP
Source LineData.ToolChange
Dest SumMsg.Data.ToolChange

Line Loop for LNi

Length 1 | Length 1 | Length 1 | Length 1

Equipment Data Summary Message 11 - Config Option 1

COP
Source LineData.Option01
Dest SumMsg.Data.Opt01
Length 1

Equipment Data Summary Message 12 - Config Option 2

COP
Source LineData.Option02
Dest SumMsg.Data.Opt02
Length 1

Equipment Data Summary Message 13 - Config Option 3

COP
Source LineData.Option03
Dest SumMsg.Data.Opt03
Length 1

Equipment Data Summary Message 14 - Config Option 4

COP
Source LineData.Option04
Dest SumMsg.Data.Opt04
Length 1

Equipment Data Summary Message 15 - Config Option 5

COP
Source LineData.Option05
Dest SumMsg.Data.Opt05
Length 1

Equipment Data Summary Message Time Total

MOVE
Source LineCfg.Takt
0
Dest SumMsg.Data.Cycle.Time
0

Equipment Data Summary Message Count Total

MOVE
Source LineData.Cycle.Count
0
Dest SumMsg.Data.Cycle.Count
0

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OR% - All Downtime

12

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

RingBuffer Modulo Result

MOD
Source A LNi
29
Source B 2
Dest BufferID
1

RingBuffer Modulo Result

EQ
Source A BufferID
1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01 (XIO)

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Line Loop for LNi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OR% - All Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

13

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend
U

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work
/

Schedule New Hour
ScheduleNewHour

Local Copy of Line Array Tag for LNi Work Time Ended
LineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 12
Dest SumMsg.Header.DataType
13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source LNi
29
Dest SumMsg.Header.ID
3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source LineData.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source LineData.EdtOR.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source LineData.EdtOR.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source LineData.LdtOR.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source LineData.LdtOR.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP
Source LineData.Full
Dest SumMsg.Data.Full

Equipment Data Summary Message 7 - Short

COP
Source LineData.Short
Dest SumMsg.Data.Short

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP
Source LineData.ProdDelay
Dest SumMsg.Data.ProdDelay

Equipment Data Summary Message 10 - Tool Change (ADC)

COP
Source LineData.ToolChange
Dest SumMsg.Data.ToolChange

Line Loop for LNi

Length 1 | Length 1 | Length 1 | Length 1

Equipment Data Summary Message 11 - Config Option 1

COP
Source LineData.Option01
Dest SumMsg.Data.Opt01
Length 1

Equipment Data Summary Message 12 - Config Option 2

COP
Source LineData.Option02
Dest SumMsg.Data.Opt02
Length 1

Equipment Data Summary Message 13 - Config Option 3

COP
Source LineData.Option03
Dest SumMsg.Data.Opt03
Length 1

Equipment Data Summary Message 14 - Config Option 4

COP
Source LineData.Option04
Dest SumMsg.Data.Opt04
Length 1

Equipment Data Summary Message 15 - Config Option 5

COP
Source LineData.Option05
Dest SumMsg.Data.Opt05
Length 1

Equipment Data Summary Message Time Total

MOVE
Source LineCfg.Takt
0
Dest SumMsg.Data.Cycle.Time
0

Equipment Data Summary Message Count Total

MOVE
Source LineData.Cycle.Count
0
Dest SumMsg.Data.Cycle.Count
0

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OA% - InDemand Downtime

14

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

RingBuffer Modulo Result

MOD
Source A LNi
29
Source B 2
Dest BufferID
1

RingBuffer Modulo Result

EQ
Source A BufferID
1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Line Loop for LNi

Copy Line working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

save updates to Line.CFG from new Equipment Callout messages

15

Equipment Data Collection Lines and Equipment Array
COP
Source LineCfg
Dest EDC.Line[LNi].CFG
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineCtl
Dest EDC.Line[LNi].CTL
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineData
Dest EDC.Line[LNi].Data
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineDataSTS
Dest EDC.Line[LNi].DataSTS
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineDataCTL
Dest EDC.Line[LNi].DataCTL
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineDataCTLprev
Dest EDC.Line[LNi].DataCTLprev
Length 1

Exit Routine

16

lbl_Exit
LBL

NOP

Reset Line Summary Data and Reset signals
(after Line, Group and Equipment for this LNi are updated)

17

Load Summary Ring Buffer by LNi
LineSumDataLoad[LnWDi].[LnBTi]
U

Reset Line, Group and Equip Data by LNi
LineDataReset[LnWDi].[LnBTi]
U

Check Viewing Index Limits

18

Line View Tag Index
LT
Source A _ViewLNi 9
Source B 0

Line View Tag Index
GT
Source A _ViewLNi 9
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Line View Tag Index
CLR
Dest _ViewLNi 9

Line Loop for LNi

Copy Line Instance to Viewing Tag if selected
(_ViewLine can be used to multiplex Trends and Diagnostic Spreadsheets)

19

_ViewEN

Line array Index

EQ	
Source A	LNi
	29
Source B	_ViewLNi
	9

Line View Tag
(used by View
Diagnostic
Spreadsheet)

COP	
Source	EDC.Line[LNi]
Dest	_ViewLine
Length	1

(End)

Equipment Loop for EQi

Equipment Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 130
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Equipment Loop for EQi

Skip Equipment if not Configured or Disabled
Typical Scan - only service Enabled equipment to reduce scan time
CfgLine Scan - check for new Equip Configuration - then also check if new Callout/Config message
DHF006 - Jul 17, 2017 -- changed Equip.CFG.EN to Equip.CTL.Enable and added CFG.Disabled (addition of Disabled Equipment support)
DHF014 - Dec 8, 2017 -- added Equip.CFG.Defined to CfgLine conditions to scan all Configured EQ (even if no Callout Message received)
EDC2 - Nov 15, 2022 - disable/ignore EQID 0000
EDC2 - Dec 5, 2022 - block CFG.Disable branch to allow Overview display of disabled EQID

2

Config Check Line LNi = CfgLNi: CfgLine
Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.Enable
lbl_Exit (JMP)

Config Check Line LNi = CfgLNi: CfgLine
Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.Enable
Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CFG.Defined
Equipment Data Buffer Lines
EQ
Source A: EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0] ??
Source B: 0

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CFG.Disable
AFI

Line array Index
EQ
Source A: LNi 29
Source B: 0

Equipment array Index
EQ
Source A: EQi 12
Source B: 0

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State
FLL
Source: 0
Dest: EquipStatus01.State
Length: 1

Local Copy of Equip Array Status Tag for LNi, EQi State = Gray - Unconfigured
EquipStatus01.State.NoCFG (L)

Server Interface Status Lines and Equipment Array
COP
Source: EquipStatus01.State
Dest: EDC_Status01.Line[LNi].Equip[EQi].State
Length: 1

Equipment Loop for EQi

3

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

Local Copy of Equip Array Tag for LNi, EQi

COP
Source EDC.Line[LNi].Equip[EQi]
Dest Equip
Length 1

Local Copy of Equip Array Status Tag for LNi, EQi

COP
Source EDC_Status01.Line[LNi].Equip[EQi]
Dest EquipStatus01
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status02.Line[LNi].Equip[EQi]
Dest EquipStatus02
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status03.Line[LNi].Equip[EQi]
Dest EquipStatus03
Length 1

4

Calculate Equipment ID for Callout Table Lookup

Equipment ID

CPT
Dest EQID
99
Expression LNi*100+EQi

Line array Index

LIMIT
Low Limit 0
Test LNi
29
High Limit CFG.CalloutLN.mi
<EDC.CFG.CalloutLN.mi>
28

Equipment array Index

LIMIT
Low Limit 0
Test EQi
12
High Limit CFG.CalloutEQ.mi
<EDC.CFG.CalloutEQ.mi>
99

Calulated EQID Index within array limits
EQID_IndexOK

5

Check for good Event Looklup Index for EDC_Event EvLKi

Equipment-Event Index for EDC_Event Lookup

CLR
Dest EvLKi
267

Local Copy of Equip Array Tag for LNi, EQi Equipment-Event Index for Lookup Tables

LIMIT
Low Limit 0
Test Equip.EventCTL.EvLKi
164
High Limit EDC_Event.CFG.EQ.mi
2999

Equipment-Event Index for EDC_Event Lookup

MOVE
Source Equip.EventCTL.EvLKi
164
Dest EvLKi
267

Event Lookup Index within array limits
EvLKi_IndexOK

Equipment Loop for EQi

Check for EQID match of Event Looklup Index for EDC_Event EvLKi

6

Event Lookup Index EQID Match EvLKi_EQIDmatch (U)

Calulated EQID Index within array limits EQID_IndexOK

Event Lookup Index within array limits EvLKi_IndexOK

Event Controls Equipment Fragment Statistics Array

EQ
Source A: EDC_Event.CTL.Equip[EvLKi].EQID (1916)
Source B: EQID (99)

Event Lookup Index EQID Match EvLKi_EQIDmatch (L)

If EQID mismatch, search for EQID in Event Lookup and update if found
(ServerInterface sets EvLKi during Event Download -- this should never be needed, but it provides a dynamic repair if something happens to index.)

7

Calulated EQID Index within array limits EQID_IndexOK

Event Lookup Index EQID Match EvLKi_EQIDmatch (/)

Event Configuration EQID Search

MOVE
Source: EDC_Event.CTL.EqQty (272)
Dest: EventEqFSC.LEN (262)

Event Configuration EQID Search

GT
Source A: EventEqFSC.LEN (262)
Source B: EDC_Event.CFG.EQ.n (3000)

Event Configuration EQID Search

MOVE
Source: EDC_Event.CFG.EQ.n (3000)
Dest: EventEqFSC.LEN (262)

Event Configuration EQID Search
EventEqFSC (RES)

Event Configuration EQID Search

MOVE
Source: 0
Dest: EventEqFSC.POS (0)

Event Configuration EQID Search

FSC
Control: EventEqFSC
Length: 262
Position: 0
Mode: ALL
Expression: EDC_Event.CTL.Equip[EventEqFSC.POS].EQID=EQID
EN / DN / ER

Event Configuration EQID Search
EventEqFSC.FD

Equipment-Event Index for EDC_Event Lookup

MOVE
Source: EventEqFSC.POS (0)
Dest: EvLKi (267)

Local Copy of Equip Array Tag for LNi, EQi Equipment-Event Index for Lookup Tables

MOVE
Source: EvLKi (267)
Dest: Equip.EventCTL.EvLKi (164)

Event Lookup Index EQID Match EvLKi_EQIDmatch (L)

Event Lookup Index Found EvLKi_Search

Equipment Loop for EQi

Callout Message
also Enables Configuration for new Equipment on the CfgLNi - one Line per scan
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL31 - during Full Download, block Callout until Equipment Defined or Eqiupment Configuration is complete -- prevents EqDriver from self-configuring and sending data
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

8

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable
/

Equipment Data Collection Download Started - Block Equipment Drivers until Equipment Config Complete
EDC.ServerCfg.Ctl.DownloadBlockEqDriver
/

Local Copy of Equip Array Tag for LNi, EQi Equipment Defined (from Server Config)
Equip.CFG.Defined

Receive Equipment Callout Unsolicited Message
JSR
Routine Name _131_EquipCalloutRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
Length CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>

Server Config or EqDriver Callout Message can Enable Equipment
EDC2 WL31 - during Full Config Download, Equip.CTL.Config is blocked until Equipment Configuration is complete -- along with Disable status

9

Local Copy of Equip Array Tag for LNi, EQi Equipment Defined (from Server Config)
Equip.CFG.Defined

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable
/

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Driver Enabled (from Driver Callout Msg)
Equip.CFG.DriverEN

Equipment Loop for EQi

Update Equipment Loop Index to include this Equipment
(should not need this, but it fixes any Loop configuration problems)

also determine the last Enabled EQi for the Line

10

Config Check Line
LNi = CfgLNi
CfgLine

Equipment array Index
GT
Source A: EQi 12
Source B: LineCfg.EQ.ei 99

Equipment array Index
LE
Source A: EQi 12
Source B: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Local Copy of Line Array Tag for LNi Array End Index
MOVE
Source: EQi 12
Dest: LineCfg.EQ.ei 99

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Equipment array Index
GT
Source A: EQi 12
Source B: LineCfg_EqENei_Temp 67

Equipment array Index
LE
Source A: EQi 12
Source B: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Temp Storage of LineCfg.EQ.ENei
MOVE
Source: EQi 12
Dest: LineCfg_EqENei_Temp 67

DHF006 - added Equip.CTL.Config and Equip.CTL.Enable
(addition of Disabled Equipment support - define CTL bits here after updating Callout status on previous rung)

11

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Calculate Equipment Data and Operational Availability (OA)
JSR
Routine Name: _140_EquipData

Equipment Loop for EQi

DHF006 - CTL.DisableEvents set by configuration to refresh Event Display to clear any existing no-Display Events

12

Local Copy of Equip Array Tag for LNi, EQi Update Events Table
Equip.EventCTL.Update

Local Copy of Equip Array Tag for LNi, EQi Equipment Events Disabled (clear existing Events in Gateway)
Equip.CTL.DisableEvents

S:FS

Update Equip Events Table for HMI Displays

JSR
Routine Name _135_EquipEvents

Local Copy of Equip Array Tag for LNi, EQi Update Events Table
Equip.EventCTL.Update
U

Local Copy of Equip Array Tag for LNi, EQi Equipment Events Disabled (clear existing Events in Gateway)
Equip.CTL.DisableEvents
U

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

OA% - InDemand Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

13

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data
Equip.DataCTL.CycleSend

Schedule New Hour
ScheduleNewHour

Local Copy of Equip Array Tag for LNi, EQi Work Time Ended
Equip.DataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE	
Source	1
Dest	SumMsg.Header.DataType
	13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE	
Source	EQID
	99
Dest	SumMsg.Header.ID
	3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP	
Source	Equip.Data.Work
Dest	SumMsg.Data.Work
Length	1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP	
Source	Equip.Data.EdtOA.Maint
Dest	SumMsg.Data.EdtMaint
Length	1

Equipment Data Summary Message 3 - Equip Downtime Production

COP	
Source	Equip.Data.EdtOA.Prod
Dest	SumMsg.Data.EdtProd
Length	1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP	
Source	Equip.Data.LdtOA.Maint
Dest	SumMsg.Data.LdtMaint
Length	1

Equipment Data Summary Message 5 - Line Downtime Production

COP	
Source	Equip.Data.LdtOA.Prod
Dest	SumMsg.Data.LdtProd
Length	1

Equipment Data Summary Message 6 -Full

COP

Equipment Data Summary Message 7 - Short

COP

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP

Equipment Data Summary Message 10 - Tool Change (ADC)

COP

Equipment Loop for EQi

Source	Equip.Data.Full
Dest	SumMsg.Data.Full
Length	1

Source	Equip.Data.Short
Dest	SumMsg.Data.Short
Length	1

Source	Equip.Data.ProdDelay
Dest	SumMsg.Data.ProdDelay
Length	1

Source	Equip.Data.ToolChange
Dest	SumMsg.Data.ToolChange
Length	1

Equipment Data Summary Message 11 - Config Option 1

COP

Source	Equip.Data.Option01
Dest	SumMsg.Data.Opt01
Length	1

Equipment Data Summary Message 12 - Config Option 2

COP

Source	Equip.Data.Option02
Dest	SumMsg.Data.Opt02
Length	1

Equipment Data Summary Message 13 - Config Option 3

COP

Source	Equip.Data.Option03
Dest	SumMsg.Data.Opt03
Length	1

Equipment Data Summary Message 14 - Config Option 4

COP

Source	Equip.Data.Option04
Dest	SumMsg.Data.Opt04
Length	1

Equipment Data Summary Message 15 - Config Option 5

COP

Source	Equip.Data.Option05
Dest	SumMsg.Data.Opt05
Length	1

Equipment Data Summary Message Time Total

MOVE

Source	Equip.CFG.Takt 56000
Dest	SumMsg.Data.Cycle.Time 0

Equipment Data Summary Message Count Total

MOVE

Source	Equip.Data.Cycle.Count 420
Dest	SumMsg.Data.Cycle.Count 0

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers

OR% - All Downtime

14

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

ADD
Source A LNi 29
Source B EQi 12
Dest LnEqSum 35

RingBuffer Modulo Result

MOD
Source A LnEqSum 35
Source B 2
Dest BufferID 1

RingBuffer Modulo Result

EQ
Source A BufferID 1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01 /

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

OR% - All Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

15

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data
Equip.DataCTL.CycleSend

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data
Equip.DataCTL.CycleSend

U

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

/

Schedule New Hour
ScheduleNewHour

Local Copy of Equip Array Tag for LNi, EQi Work Time Ended
Equip.DataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 11
Dest SumMsg.Header.DataType
13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source EQID
99
Dest SumMsg.Header.ID
3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source Equip.Data.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source Equip.Data.EdtOR.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source Equip.Data.EdtOR.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source Equip.Data.LdtOR.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source Equip.Data.LdtOR.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP

Equipment Data Summary Message 7 - Short

COP

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP

Equipment Data Summary Message 10 - Tool Change (ADC)

COP

Equipment Loop for EQi

Source	Equip.Data.Full
Dest	SumMsg.Data.Full
Length	1

Source	Equip.Data.Short
Dest	SumMsg.Data.Short
Length	1

Source	Equip.Data.ProdDelay
Dest	SumMsg.Data.ProdDelay
Length	1

Source	Equip.Data.ToolChange
Dest	SumMsg.Data.ToolChange
Length	1

Equipment Data Summary Message 11 - Config Option 1

COP

Source	Equip.Data.Option01
Dest	SumMsg.Data.Opt01
Length	1

Equipment Data Summary Message 12 - Config Option 2

COP

Source	Equip.Data.Option02
Dest	SumMsg.Data.Opt02
Length	1

Equipment Data Summary Message 13 - Config Option 3

COP

Source	Equip.Data.Option03
Dest	SumMsg.Data.Opt03
Length	1

Equipment Data Summary Message 14 - Config Option 4

COP

Source	Equip.Data.Option04
Dest	SumMsg.Data.Opt04
Length	1

Equipment Data Summary Message 15 - Config Option 5

COP

Source	Equip.Data.Option05
Dest	SumMsg.Data.Opt05
Length	1

Equipment Data Summary Message Time Total

MOVE

Source	Equip.CFG.Takt 56000
Dest	SumMsg.Data.Cycle.Time 0

Equipment Data Summary Message Count Total

MOVE

Source	Equip.Data.Cycle.Count 420
Dest	SumMsg.Data.Cycle.Count 0

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers

OA% - InDemand Downtime

16

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

ADD
Source A LNi 29
Source B EQi 12
Dest LnEqSum 35

RingBuffer Modulo Result

MOD
Source A LnEqSum 35
Source B 2
Dest BufferID 1

RingBuffer Modulo Result

EQ
Source A BufferID 1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01 /

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Equipment Comm Status
-- Set Default State = Configured if no State value (don't run Equipment Timer)
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

17

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

COP
Source EquipStatus01.State
Dest EquipStatus01.StateVal
Length 1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

LE
Source A EquipStatus01.StateVal 16#0000_0012
Source B 1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State

FLL
Source 0
Dest EquipStatus01.State
Length 1

Local Copy of Equip Array Status Tag for LNi, EQi State = White - Configured (never received any Data Requests)
EquipStatus01.State.CFG
L

Equipment Loop for EQi

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

18

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

COP	
Source	EquipStatus01.State
Dest	EquipStatus01.StateVal
Length	1

Equipment Data Collection Lines and Equipment Array

COP	
Source	Equip
Dest	EDC.Line[LNi].Equip[EQi]
Length	1

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[LNi].Equip[EQi]
Length	1

Event Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus02
Dest	EDC_Status02.Line[LNi].Equip[EQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[LNi].Equip[EQi]
Length	1

Exit Routine

19

lbl_Exit
LBL

NOP

Check Viewing Index Limits

20

Equipment View Tag Index

LT	
Source A	_ViewEQi 18
Source B	0

Equipment View Tag Index

GT	
Source A	_ViewEQi 18
Source B	CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Equipment View Tag Index

CLR	
Dest	_ViewEQi 18

Equipment Loop for EQi

Copy Equipment Instance to Viewing Tag if selected
(_ViewEquip can be used to multiplex Trends and Diagnostic Spreadsheets)

21

_ViewEN

Line array Index

EQ	
Source A	LNi
	29
Source B	_ViewLNi
	9

Equipment array Index

EQ	
Source A	EQi
	12
Source B	_ViewEQi
	18

Equipment View Tag (used by View Diagnostic Spreadsheet)

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	_ViewEquip
Length	1

(End)

Receive Equipment Callout
Unsolicited Message

Receive Equipment Callout/Status Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 131
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)
EDC2 WL18 - Block Transaction TImer if Fault Message not enabled for PdM-only without EqDriver

2

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.CalloutMsg.TransTMR.PRE 999999999

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.CalloutMsg.TransTMR
Preset 999999999
Accum 308
EN
DN

Receive Equipment Callout
Unsolicited Message
Transaction Timeout
Callout Message not Received
Gateway-generated Message clear stale Faults and report Callout Message Error

3

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A: Equip.CTL.CalloutMsg.TransTMR.ACC
308
Source B: CFG.LiveMsgTimeout
<EDC.CFG.LiveMsgTimeout>
150000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Rx Msg ER
EquipStatus01.Update.CalloutRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: Equip.CTL.CalloutMsg
PerfTrans: EquipStatus02.CalloutRx.ER

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 1004
Input Par: Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: 0

Equipment Data Callout/Status Message

FLL
Source: 0
Dest: CalloutMsg
Length: 1

Equipment Data Callout/Status Message Transaction Time Stamp

COP
Source: CLX.TransTime
<EDC.CLX.TransTime>
Dest: CalloutMsg.TransTime
Length: 1

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

CPT
Dest: CalloutMsg.Header.EQID
1916
Expression: LNi*100+EQi

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source: 0
Dest: CalloutMsg.Header.PacketID
7

Equipment Data Callout/Status Message Gateway Message Error (from EqDriver)
CalloutMsg.PreDef._31
L

Equipment Data Callout/Status Message Alarm Status is ON (from EqDriver Alarm Mask)
CalloutMsg.PreDef._07
L

lbl_RxLoad
JMP

Receive Equipment Callout
Unsolicited Message

Transaction Timeout
Cleanup Callout Messgae ER if EQ removed from Configuration
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

4

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable
/

Equipment Data Callout/Status Message Gateway Message Error (from EqDriver)
CalloutMsg.PreDef._31

Equipment Data Callout/Status Message
FLL
Source 0
Dest CalloutMsg
Length 1

Equipment Data Callout/Status Message Gateway Message Error (from EqDriver)
CalloutMsg.PreDef._31
U

lbl_RxLoad
JMP

No new data in Rx Buffer
Header Transaction Number =0

5

Equipment Data Buffer Lines
EQ
Source A EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
??
Source B 0

lbl_Exit
JMP

Copy Rx Buffer to local registers
DINT Array to UDT

6

Equipment Data Callout/Status Message
CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
Dest CalloutMsg
Length 1

Receive Equipment Callout
Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

7

EQ
Source A CalloutMsg.Header.TransNum
0
Source B 0

EQ
Source A CalloutMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

New data in Rx Buffer
Initialize Transaction timing

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.EN

8

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.CalloutMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.CalloutMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CalloutMsg
PerfTrans EquipStatus02.CalloutRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 1000
Input Par Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

9

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ
Source A CalloutMsg.Header.TransNum
0
Source B 0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

NE
Source A CalloutMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.CalloutMsg.PartialTMR.PRE
10000

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.CalloutMsg.PartialTMR
Preset 10000
Accum 0
EN
DN

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE
Source A CalloutMsg.Header.TransNum
0
Source B 0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A CalloutMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

10

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.DN
/

lbl_Exit
JMP

Receive Equipment Callout
Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

11

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CalloutMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Rx Msg ER
EquipStatus01.Update.CalloutRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1005
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

lbl_NextTrans
JMP

Receive Equipment Callout
Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Callout table if mismatch

12

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE
Source A CalloutMsg.Header.TransNum
0
Source B 0
0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

NE
Source A CalloutMsg.Footer.TransNum
0
Source B 0
0

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE
Source A CalloutMsg.Header.TransNum
0
Source B CalloutMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.CalloutMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Rx Msg ER

EquipStatus01.Update.CalloutRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CalloutMsg
PerfTrans EquipStatus02.CalloutRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 1006
Input Par Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CalloutMsg.Header.TransNum

lbl_NextTrans
JMP

Receive Equipment Callout
Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

13

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum 0
Source B	0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Footer.TransNum 0
Source B	0

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ	
Source A	CalloutMsg.Header.TransNum 0
Source B	CalloutMsg.Footer.TransNum 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.CalloutMsg.TransRcvd

L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Callout Rx DN

EquipStatus01.Update.CalloutRxOK

L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.DN

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1001
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Callout table if mismatch

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID
	85
Expression	LNi*100+EQi

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	CalloutMsg.Header.EQID
	1916
Source B	ExpectedEQID
	85

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Equipment ID Mismatch ER
EquipStatus01.Update.CalloutRxEqIdER
(L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.EQID
	1916
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1007
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

lbl_NextTrans
(JMP)

Receive Equipment Callout
Unsolicited Message

Check new message for duplicate Transaction Number
Load Callout table but indicate issue in performance/diagnostic

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ	
Source A	CalloutMsg.Header.TransNum 0
Source B	Equip.CTL.CalloutMsg.TransNumLast 6451

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Callout Rx DN
EquipStatus01.Update.CalloutRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.CalloutMsg.TransNumLast 6451
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.KA

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1002
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load Callout table but indicate issue in performance/diagnostic

16

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum 0
Source B	Equip.CTL.CalloutMsg.TransNumLast 6451

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum 0
Source B	Equip.CTL.CalloutMsg.TransNumNext 6452

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Transaction Number Sequence ER
EquipStatus01.Update.CalloutRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.CalloutMsg.TransNumNext 6452
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1009
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message
Load new message in Callout Table

17

lbl_RxLoad
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

Load Equipment Callout Message into Callout Table
JSR
Routine Name _132_EquipCalloutRxLoad

Prepare for next message

18

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number
MOVE
Source CalloutMsg.Header.TransNum 0
Dest Equip.CTL.CalloutMsg.TransNumLast 6451

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
ADD
Source A CalloutMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.CalloutMsg.TransNumNext 6452

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
LT
Source A Equip.CTL.CalloutMsg.TransNumNext 6452
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
GT
Source A Equip.CTL.CalloutMsg.TransNumNext 6452
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
MOVE
Source 1
Dest Equip.CTL.CalloutMsg.TransNumNext 6452

Receive Equipment Callout
Unsolicited Message
Exit path when Transaction Reset or Received

lbl_NextTrans

19 LBL — NOP

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)
EDC2 - v2.01 add Event Configuration for DtOwner and DtType
(send Response with Event Masks if Server Download All complete - during Download, EqDriver will use existing Masks)

20

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CalloutMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CalloutMsg.TransReset — U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd — U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout — U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR — RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.CalloutMsg.TransTMR — RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.CalloutMsg.TransLast
0

Equipment Data Callout/Status Message

FLL
Source 0
Dest CalloutRspMsg
Length 1

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Equipment Data Collection Response Download All Started
EDC.ServerCfg.Ctl.DownloadAllStarted — /

Create Callout Response Message with Event Mask

JSR
Routine Name _134_EquipCalloutRxResponse

Equipment Data Lines

COP
Source CalloutMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Callout
Length 1

Equipment Data Buffer Lines

COP
Source CalloutRspMsg
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
Length CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>

Exit path when no Message, Partial Message or Corrupt Message

lbl_Exit

21 LBL — NOP

Receive Equipment Callout
Unsolicited Message

(End)

Load Equipment Callout Message into Callout Table

Load Equipment Callout Message into Callout Table

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 132
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Convert Header EQID to LNi and EQi

2

MOD
Source A CalloutMsg.Header.EQID 1916
Source B 100
Dest HeaderEQi 85

CPT
Dest HeaderLNi 0
Expression (CalloutMsg.Header.EQID - HeaderEQi) / 100

Load Equipment Callout Message into Callout Table

Equipment ID outside Callout Table array limits
Log Transaction and Exit

3

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

LT
Source A HeaderLNi 0
Source B 0

GT
Source A HeaderLNi 0
Source B CFG.CalloutLN.mi <EDC.CFG.CalloutLN.mi> 28

LT
Source A HeaderEQi 85
Source B 0

GT
Source A HeaderEQi 85
Source B CFG.CalloutEQ.mi <EDC.CFG.CalloutEQ.mi> 99

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Header.EQID 1916
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Header.PacketID 7
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 1013
Input Par Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CalloutMsg.Header.TransNum

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CalloutMsg
PerfTrans EquipStatus02.CalloutRx.ER

lbl_Exit
JMP

Load Equipment Callout Message into Callout Table

v2.0 - Callout Msg Version determined by Footer.Spare0=v1.6 (old PreDef layout), 2=v2.0 (new PreDef layout)
remap v1.6 to v2.0 PreDefined layout

July 25, 2022 -- blocked v1.6 remap - was clearing No Event Display (bit 15) - can't do this for v1.6 drivers until configuration is updated

4

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)

EQ
Source A CalloutMsg.Footer.Spare 0
Source B 2

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Remap Predefined

JSR
Routine Name _133_EquipCalloutRxMapV1

Load Equipment Callout Message into Callout Table

Load Message into Callout Table
Set Update bits for Equip Event Table, HMI Overview Event Table and HMI Zones Event Tables
(these are used to indicate Equipment Event Table updates needed when new Callout Messages)

v1.3 Support Multiple Fault/Callout Packets
If Packet 0, Update Callout with PreDefined & EqDefined, otherwise just Load PreDefined

DN014 Health Tool Status Download from Server
Callout bits 19 (alarm) & 20 (fault) from Tool EqDriver are blocked to allow Server to control status
Callout bit 6 (EQ Fault) is turned ON when the Health Fault is ON

5

COP
Source CalloutMsg.PreDef
Dest TempDINT
Length 1

Callout Table (LNi, EQi)
MVM
Source TempDINT
16#0000_0023
Mask 16#ffe7_ffff
Dest EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef
16#0000_0024

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet
LIMIT
Low Limit 0
Test CalloutMsg.Header.PacketID
7
High Limit CFG.CalloutPKT.mi
<EDC.CFG.CalloutPKT.mi>
7

MUL
Source A CalloutMsg.Header.PacketID
7
Source B CFG.CalloutEqDef.n
<EDC.CFG.CalloutEqDef.n>
8
Dest PktWDi
24

Callout Table (LNi, EQi)
COP
Source CalloutMsg.EqDef.Word[0]
Dest EDC_EqCallout[HeaderLNi,HeaderEQi].EqDef[PktWDi]
Length CFG.CalloutEqDef.n
<EDC.CFG.CalloutEqDef.n>

Callout Table (LNi, EQi)
EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef.19
AFI
Callout Table (LNi, EQi)
EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef.20

Callout Table (LNi, EQi)
EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef.6
L

Local Copy of Equip Array Tag for LNi, EQi Update Events Table
Equip.EventCTL.Update
L

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Overview Display
Equip.EventCTL.UpdateOview
L

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Zone Displays
FLL
Source -1
Dest Equip.EventCTL.UpdateZone
Length 1

Load Equipment Callout Message into Callout Table

Set Line/Equipment Configured to enable service of Fault, Quality, Trend Messages
Update Line/Equipment scan Indexes to include this Equipment
DHF006 - changed CFG.EN to CFG.DriverEN (addition of Disabled Equipment support)
EDC2 WL08 - add Cycle Data Message

6

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

Equipment Data Callout/Status Message Driver Enabled
CalloutMsg.Header.DriverEN

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Local Copy of Equip Array Tag for LNi, EQi Driver Enabled (from Driver Callout Msg)
Equip.CFG.DriverEN

Line array Index

GT	
Source A	LNi
	29
Source B	CFG.LN.ei
	<EDC.CFG.LN.ei>
	28

Line array Index

LE	
Source A	LNi
	29
Source B	CFG.LN.mi
	<EDC.CFG.LN.mi>
	28

EDC Config Array End Index

MOVE	
Source	LNi
	29
Dest	CFG.LN.ei
	<EDC.CFG.LN.ei>
	28

Equipment array Index

GT	
Source A	EQi
	12
Source B	LineCfg.EQ.ei
	99

Equipment array Index

LE	
Source A	EQi
	12
Source B	CFG.EQ.mi
	<EDC.CFG.EQ.mi>
	99

Local Copy of Line Array Tag for LNi Array End Index

MOVE	
Source	EQi
	12
Dest	LineCfg.EQ.ei
	99

Equipment Data Callout/Status Message Fault Message Enabled
CalloutMsg.Header.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Equipment Data Callout/Status Message Quality Message Enabled
CalloutMsg.Header.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

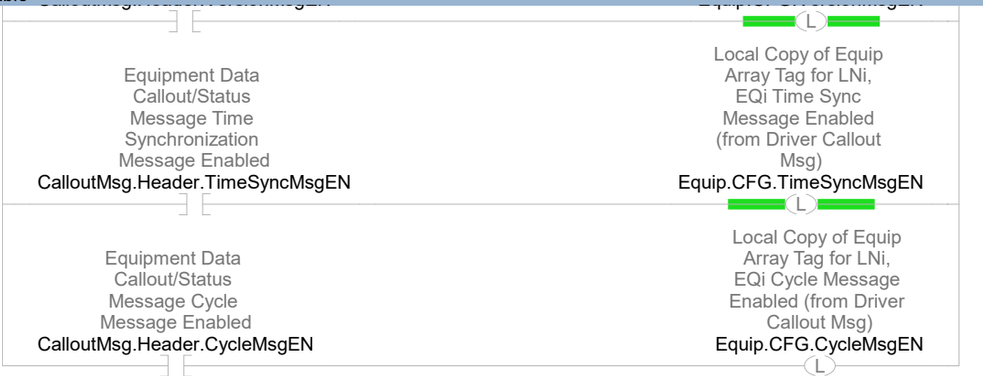
Equipment Data Callout/Status Message Trend Message Enabled
CalloutMsg.Header.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Equipment Data Callout/Status Message Version Message Enabled
CalloutMsg.Header.VersionMsgEN

Local Copy of Equip Array Tag for LNi, EQi Version Message Enabled (from Driver Callout Msg)
Equip.CFG.VersionMsgEN

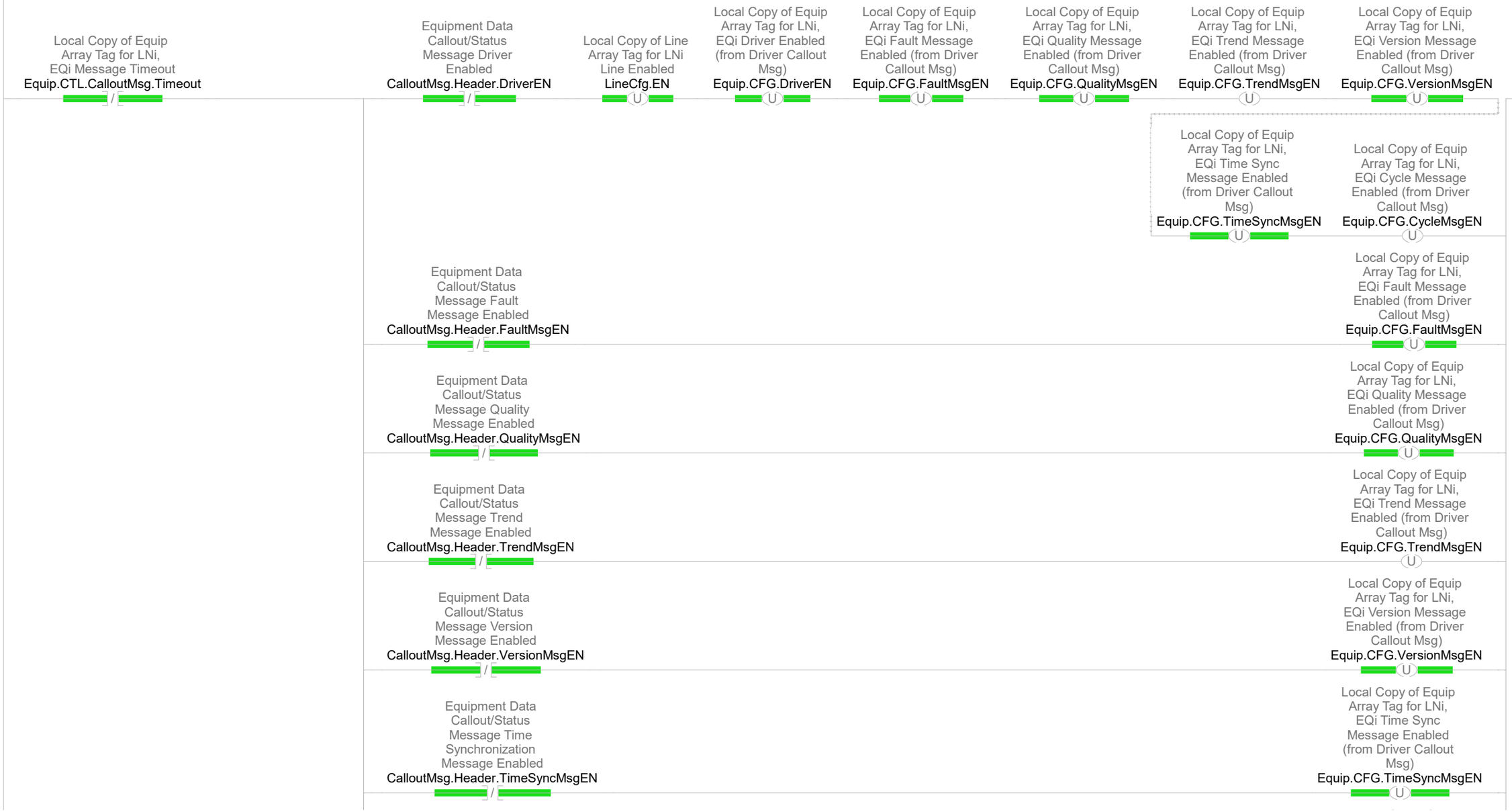
Load Equipment Callout Message into Callout Table



Load Equipment Callout Message into Callout Table

Reset Line/Equipment Configured to enable service of Fault, Quality, Trend Messages
DHF006 - changed CFG.EN to CFG.DriverEN (addition of Disabled Equipment support)
EDC2 WL08 - add Cycle Data Message

7



Load Equipment Callout Message into Callout Table

Equipment Data
Callout/Status
Message Cycle
Message Enabled
CalloutMsg.Header.CycleMsgEN

Local Copy of Equip
Array Tag for LNi,
EQi Cycle Message
Enabled (from Driver
Callout Msg)
Equip.CFG.CycleMsgEN
U

Skip Option Update if EQID Event Lookup Index is not good

Event Lookup Index
EQID Match
EvLKi_EQIDmatch

lbl_OptionsDone
JMP

8

Packet Index

MOVE
Source CalloutMsg.Header.PacketID
7
Dest PKTi
3

9

Skip Option Update if EQID Event Option 1-5 are not configured
faster to skip over AOI than solve false
(fwiw - did not see Callout scan time improvement with AOI skips)

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi]

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi]

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi]

10

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi]

lbl_OptionsDone
JMP

Skip Option Update if EQID Event Option not configured
faster to skip over AOI than solve false

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]

lbl_Option2
JMP

11

Load Equipment Callout Message into Callout Table

Update Option 1

12

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt01
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[1] 5796936
Source B	1
Dest	KpiOptionDiag[1] 5796936

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

13

Event KPI Options Equipment Event KPI Options

lbl_Option2 LBL — EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi] /

lbl_Option3 JMP

Update Option 2

14

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt02
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[2] 1041056
Source B	1
Dest	KpiOptionDiag[2] 1041056

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

15

Event KPI Options Equipment Event KPI Options

lbl_Option3 LBL — EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi] /

lbl_Option4 JMP

Load Equipment Callout Message into Callout Table

Update Option 3

16

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt03
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[3] 1295
Source B	1
Dest	KpiOptionDiag[3] 1295

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

17

Event KPI Options Equipment Event KPI Options

lbl_Option4 LBL — EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi] / — lbl_Option5 JMP

Update Option 4

18

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt04
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[4] 1295
Source B	1
Dest	KpiOptionDiag[4] 1295

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

19

Event KPI Options Equipment Event KPI Options

lbl_Option5 LBL — EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi] / — lbl_OptionsDone JMP

Load Equipment Callout Message into Callout Table

Update Option 5

Use the Option Masks to determine Callout Data Packet Option Status

```
aoiEDC_SumOptionStatus
aoiEDC_SumOptionStat...  aoiEDC_SumOptionStatus (...)
EventData                CalloutMsg.EqDef.Word
OptMask                  EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data
EventOpt                 EDC_EventOpt.Equip[EvLKi].Opt05
PKTi                     PKTi
```

KPI Option Diagnostic

```
ADD
Source A   KpiOptionDiag[5]
           1295
Source B   1
Dest       KpiOptionDiag[5]
           1295
```

20

Option Update Done

lbl_OptionsDone

LBL

21

NOP

Callout Table Write -- Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

```
MOVE
Source   CalloutMsg.Header.EQID
         1916
Dest     Trans.Opt[1]
         <EDC.Trans.Opt[1]>
         -1
```

Transaction Log Control SBR External Load -- Optional Parameters

```
MOVE
Source   CalloutMsg.Header.PacketID
         7
Dest     Trans.Opt[2]
         <EDC.Trans.Opt[2]>
         -1
```

Log Transactions

```
JSR
Routine Name   _901_TransLogSBR
Input Par      1011
Input Par      Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par      LNi
Input Par      EQi
Input Par      CalloutMsg.Header.TransNum
```

22

Exit Routine

lbl_Exit

LBL

23

NOP

(End)

Remap Predefined

v2.0 - Callout Msg Version determined by Header.Cmd 0=v1.6 (old PreDef layout), 2=v2.0 (new PreDef layout)
remap v1.6 to v2.0 PreDefined layout

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 133
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

No v1 mapping needed in new EDC2 deployments
(don't want to remap accidentally if EqDriver is missing version)

2

NOP

(End)

Create Callout Response Message with Event Mask

EDC2 - v2.01 add Event Configuration for DtOwner and DtType
Create Equipment Callout Response Message with one Event Mask
Equipment Driver will use the Masks to Ignore Events, set DT Owner (Maint/Prod) and DT Type (Fault, Alarm, Alert)

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 134
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Exit if no Event Lookup Index EQID Match

2

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

lbl_Exit
JMP

Initialize Callout Response Message for PKTi
Response Message always has same PKTi as Callout Message
(no point in sending unused Packet configurations)

3

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source CalloutMsg.Header.PacketID 7
Dest CalloutRspMsg.Header.PacketID 3

Create Callout Response Message with Event Mask

4

Load EQID & Packet ID

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE	
Source	CalloutMsg.Header.EQID 1916
Dest	CalloutRspMsg.Header.EQID 1004

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE	
Source	CalloutMsg.Header.PacketID 7
Dest	CalloutRspMsg.Header.PacketID 3

Packet Index

MOVE	
Source	CalloutMsg.Header.PacketID 7
Dest	PKTi 3

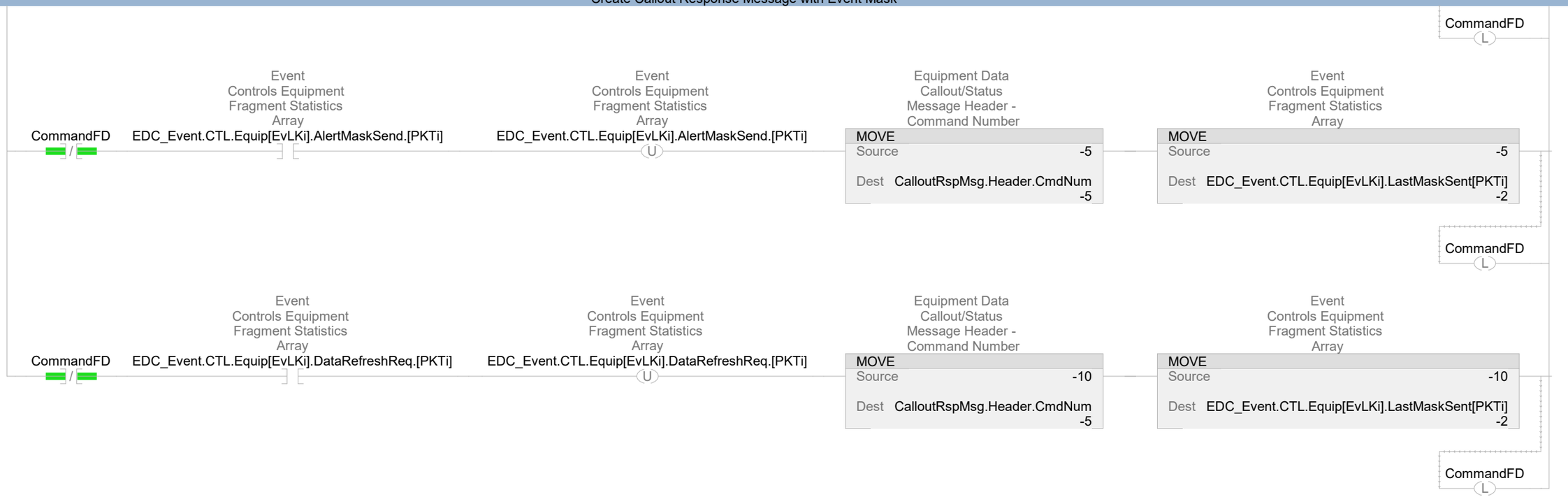
Create Callout Response Message with Event Mask

Select Packet-Command Mask for Response
Prioritize new Config items that haven't been sent
v3.0.2 Data Refresh Request Command

5

CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
U	EDC_Event.CTL.Equip[EvLKi].MaintMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].MaintMaskSend.[PKTi]	MOVE Source -1 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -1 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
				L CommandFD
CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
/	EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend.[PKTi]	MOVE Source -2 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -2 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
U				L CommandFD
CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
/	EDC_Event.CTL.Equip[EvLKi].FaultMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].FaultMaskSend.[PKTi]	MOVE Source -3 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -3 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
U				L CommandFD
CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
/	EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend.[PKTi]	MOVE Source -4 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -4 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
U				

Create Callout Response Message with Event Mask



Create Callout Response Message with Event Mask

If no undelivered Mask for this Packet, send the next Mask
Commands -1 thru -5

6

CommandFD

SUB
Source A: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Source B: 1
Dest: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2

LT
Source A: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Source B: -5

GT
Source A: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Source B: -1

MOVE
Source: -1
Dest: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2

MOVE
Source: Equipment Data Callout/Status Message Header - Command Number EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Dest: CalloutRspMsg.Header.CmdNum -5

7

EQ
Source A: Equipment Data Callout/Status Message Header - Command Number CalloutRspMsg.Header.CmdNum -5
Source B: -1

COP
Source: Maintenance Event Mask Downtime Owner EDC_EventMaskPreDef.Maint
Dest: Equipment Data Callout/Status Message Data - Predefined Status CalloutRspMsg.PreDef
Length: 1

COP
Source: EDC_EventMaskMaint.Equip[EvLKi].Packet[PKTi].Data[0]
Dest: Equipment Data Callout/Status Message Equipment Defined Status Words CalloutRspMsg.EqDef.Word[0]
Length: EDC_Event.CFG.DAT.n

Create Callout Response Message with Event Mask

Ignore Event Mask
Downtime Type

8

Equipment Data Callout/Status Message Header - Command Number

EQ	
Source A	CalloutRspMsg.Header.CmdNum
	-5
Source B	-2

Equipment Data Callout/Status Message Data - Predefined Status

COP	
Source	EDC_EventMaskPreDef.Ignore
Dest	CalloutRspMsg.PreDef
Length	1

Equipment Data Callout/Status Message Equipment Defined Status Words

COP	
Source	EDC_EventMaskIgnore.Equip[EvLKi].Packet[PKTi].Data[0]
Dest	CalloutRspMsg.EqDef.Word[0]
Length	EDC_Event.CFG.DAT.n

Fault Event Mask
Downtime Type

9

Equipment Data Callout/Status Message Header - Command Number

EQ	
Source A	CalloutRspMsg.Header.CmdNum
	-5
Source B	-3

Equipment Data Callout/Status Message Data - Predefined Status

COP	
Source	EDC_EventMaskPreDef.Fault
Dest	CalloutRspMsg.PreDef
Length	1

Equipment Data Callout/Status Message Equipment Defined Status Words

COP	
Source	EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[0]
Dest	CalloutRspMsg.EqDef.Word[0]
Length	EDC_Event.CFG.DAT.n

Alarm Event Mask
Downtime Type

10

Equipment Data Callout/Status Message Header - Command Number

EQ	
Source A	CalloutRspMsg.Header.CmdNum
	-5
Source B	-4

Equipment Data Callout/Status Message Data - Predefined Status

COP	
Source	EDC_EventMaskPreDef.Alarm
Dest	CalloutRspMsg.PreDef
Length	1

Equipment Data Callout/Status Message Equipment Defined Status Words

COP	
Source	EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[0]
Dest	CalloutRspMsg.EqDef.Word[0]
Length	EDC_Event.CFG.DAT.n

Create Callout Response Message with Event Mask

Alert Event Mask
Downtime Type

11

EQ
Source A: CalloutRspMsg.Header.CmdNum
-5
Source B: -5

Equipment Data Callout/Status Message Header - Command Number

COP
Source: EDC_EventMaskPreDef.Alert
Dest: CalloutRspMsg.PreDef
Length: 1

Equipment Data Callout/Status Message Data - Predefined Status

COP
Source: EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[0]
Dest: CalloutRspMsg.EqDef.Word[0]
Length: EDC_Event.CFG.DAT.n

Equipment Data Callout/Status Message Equipment Defined Status Words

Generate the Mask Download Required to indicates which Packets have more Masks to download.
The EqDriver uses this to generate Callout Messages until all Masks are downloaded
(waiting for data events to occur in each Packet is not a reliable mechanism)
v3.02 Data Refresh Request Command

12

CPT
Dest
Expression: EDC_Event.CTL.Equip[EvLKi].MaintMaskSend OR EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend OR EDC_Event.CTL.Equip[EvLKi].FaultMaskSend OR EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend OR EDC_Event.CTL.Equip[EvLKi].AlertMaskSend

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)

Create Callout Response Message with Event Mask

CalloutRspMsg.Footer.Spare
0
)R EDC_Event.CTL.Equip[EvLKi].DataRefreshReq

Exit Routine

lbl_Exit

13 LBL NOP

(End)

Update Equip Events Table for HMI Displays

Update Top-N Equipment Events
Table of Current Events and Start Time ordered by Duration (oldest first)

This method provides time-ordered list without sorting Events by duration
- Remove Existing Events from List
- Add New Events in order found until table full
- If table is full, a Remove/Add may not find Oldest New Event - it will find the Next New Event in Word, Bit order
(this is a small tradeoff to avoid sort by duration for all Events)

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 135
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

DN014 - Replace CallMsg.Header.EQID with EQID to allow Equip Event updates without Callout Msg
Calculate Equipment ID for Callout Table Lookup
Exit if EQID out of range

2

Calulated EQID Index within array limits
EQID_IndexOK
/

lbl_Exit
JMP

Update Equip Events Table for HMI Displays

Check the current Events and remove any that have turned off.
By adding/removed Events from the current Event list, the oldest events are retained, in order without sorting by Duration

Copy Callout Table to DINT array for FBC
Initialize Event Loop Pointer

DN014 - Replace CallMsg.Header.EQID with EQID to allow Equip Event updates without Callout Msg

3

Local Copy of EDC_EqCallout for this EQID

COP	
Source	EDC_EqCallout[LNi,EQi]
Dest	TempCallout[0]
Length	CFG.CalloutARR.n <EDC.CFG.CalloutARR.n>

Event Table Index

CLR	
Dest	EVi
	0

Skip Event Mask load if invalid Lookup Index default to Downtime Type of Fault

4

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

Temp Result for Array Size

MUL	
Source A	EDC_Event.CFG.PKT.n[8]
Source B	EDC_Event.CFG.DAT.n[8]
Dest	LEN 64

Equipment Event Fault Mask

FLL	
Source	-1
Dest	EquipMaskFault[0]
Length	LEN

Equipment Event Alarm Mask

FLL	
Source	0
Dest	EquipMaskAlarm[0]
Length	LEN

Equipment Event Alert Mask

FLL	
Source	0
Dest	EquipMaskAlert[0]
Length	LEN

Equipment Event No Display Mask

FLL	
Source	0
Dest	EquipMaskNoDisp[0]
Length	LEN

lbl_SkipMaskLoad
JMP

Get EqDefined Event Masks for Fault, Alarm and Alert used to determine Event Display status

5

Temp Result for Array Size

MUL	
Source A	EDC_Event.CFG.PKT.n[8]
Source B	EDC_Event.CFG.DAT.n[8]
Dest	LEN 64

Equipment Event Fault Mask

COP	
Source	EDC_EventMaskFault.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskFault[0]
Length	LEN

Equipment Event Alarm Mask

COP	
Source	EDC_EventMaskAlarm.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskAlarm[0]
Length	LEN

Equipment Event Alert Mask

COP	
Source	EDC_EventMaskAlert.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskAlert[0]
Length	LEN

Equipment Event No Display Mask

COP	
Source	EDC_EventMaskNoDisp.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskNoDisp[0]
Length	LEN

Update Equip Events Table for HMI Displays

Skip Equipment Event Mask Load

lbl_SkipMaskLoad

6 LBL — NOP

Equipment Event Display Status
- EQ InterlockOff - display only Interlock Off Event
- EQ not InterlockOff, Fault, Alarm or Alert - display no Events
- NoDisplay (Predefined.15) - display no Events (from v1.x EquDriver)
DHF006 - CTL.DisableEvents clears Events
DHF029 - Added Alert Status 5
EDC2 - v2.01 added Equip.CFG.DisableEvents to replace PreDefined.15

7

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0 /

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff /

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red)
Equip.DataCTL.Fault /

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow)
Equip.DataCTL.Alarm /

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue)
Equip.DataCTL.Alert /

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

EQ NoEventDisplay
TempCallout[0].15
AFI

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Local Copy of Equip Array Tag for LNi, EQi Equipment Event Display Disabled (from Server Config)
Equip.CFG.DisableEvents

Update Equip Events Table for HMI Displays

Exit Event Check Loop if Record is Empty

8

lbl_EventRemoveLoop — LBL

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element (Word*100) + Bit

EQ
Source A: Equip.Event[EVi].Element -1
Source B: -1

lbl_EventAdd — JMP

Check if Element Word/Bit Indexes in Range
Remove if out of Range

9

This Event local copy of Equip.Event[EVi]

COP
Source: Equip.Event[EVi]
Dest: Event
Length: 1

Display Event Removed: EventRemoved — U

This Event local copy of Equip.Event[EVi] Fault/Status Event Element Word

LIMIT
Low Limit: 0
Test: Event.Word 33
High Limit: CFG.CalloutARR.mi <EDC.CFG.CalloutARR.mi> 64

This Event local copy of Equip.Event[EVi] Fault/Status Event Element Bit

LIMIT
Low Limit: 0
Test: Event.Bit 11
High Limit: 31

Event Word Index

MOVE
Source: Event.Word 33
Dest: WDi 39

Event Bit Index

MOVE
Source: Event.Bit 11
Dest: BITi 17

EventIndexInRange

EventIndexInRange — /

lbl_EventRemove — JMP

PreDefined Event No Display
only display Fault, Alarm and Alert
v3.5 moved v2_0 inside branch; always honor NoDisp

10

Event Word Index

EQ
Source A: WDi 39
Source B: 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)

Equip.CTL.CalloutMsgV2_0

PreDefined Event Mask DtType Fault Mask

EDC_EventMaskPreDef.Fault.[BITi] — /

PreDefined Event Mask DtType Alarm Mask

EDC_EventMaskPreDef.Alarm.[BITi] — /

PreDefined Event Mask DtType Alert Mask

EDC_EventMaskPreDef.Alert.[BITi] — /

PreDefined Event Mask No Display Option

EDC_EventMaskPreDef.NoDisp.[BITi]

PreDefined No-Display Event

EventNoDisplayRemove_PreDef

Update Equip Events Table for HMI Displays

EqDefined Event No Display
only display Fault, Alarm and Alert
v3.5 moved v2_0 inside branch; always honor NoDisp

11

Event Word Index
GT
Source A WDi 39
Source B 0

Equip-Defined Event Word Index
SUB
Source A WDi 39
Source B 1
Dest EqDef_WDi 38

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Equipment Event Fault Mask
EquipMaskFault[EqDef_WDi].[BITi]

Equipment Event Alarm Mask
EquipMaskAlarm[EqDef_WDi].[BITi]

Equipment Event Alert Mask
EquipMaskAlert[EqDef_WDi].[BITi]

Equipment Event No Display Mask
EquipMaskNoDisp[EqDef_WDi].[BITi]

EqDefined No-Display Event
EventNoDisplayRemove_EqDef

Update Equip Events Table for HMI Displays

Check if Event still On
Remove if Event is not On or Equipment/Event is No Display

PreDefined Word 0, Bits 0-7 and Bit 17 are not Displayed Events
(OverallStart, ConvRunning, FixedStopPosition, FixedStopReq, TempStopReq, ProdStopReq, Fault, Alarm, Alert)
DHF029 - Added Alert to not Displayed Events

12

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

Remove No-Display Event from Event Viewers
EventNoDisplayRemove

PreDefined No-Display Event
EventNoDisplayRemove_PreDef

EqDefined No-Display Event
EventNoDisplayRemove_EqDef

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Event Word Index
NE
Source A WDi 39
Source B 0

Event Bit Index
NE
Source A BITi 17
Source B 14

Event Word Index
EQ
Source A WDi 39
Source B 0

Event Bit Index
LIMIT
Low Limit 0
Test BITi 17
High Limit 7

Event Bit Index
EQ
Source A BITi 17
Source B 17

Update Equip Events Table for HMI Displays

Remove No-Display Event from Event Viewers
EventNoDisplayRemove
/

Local Copy of EDC_EqCallout for this EQID
TempCallout[WDi].[BITi]

lbl_EventNext
JMP

Remove Event
If this isn't the last Event in the Table, shift following Events up 1 position before clearing last position

13

lbl_EventRemove
LBL

Event Table Index
LT
Source A EVi
0
Source B CFG.EV.mi
<EDC.CFG.EV.mi>
9

Temp Result for Array Size
SUB
Source A CFG.EV.mi
<EDC.CFG.EV.mi>
9
Source B EVi
0
Dest LEN
64

Local Copy of Equip Array Tag for LNi, EQi Equipment Event Table First N Fault/Status Events
COP
Source Equip.Event[EVi+1]
Dest Equip.Event[EVi]
Length LEN

Display Event Removed
EventRemoved
L

Remove Event
Clear Last Position in Table

14

Local Copy of Equip Array Tag for LNi, EQi Equipment Event Table First N Fault/Status Events
FLL
Source -1
Dest Equip.Event[CFG.EV.mi]
Length 1

Update Equip Events Table for HMI Displays

Increment Index
Block Increment if Event Deleted - Event Table Shift moves next Event to current Index
If EVi exceeds Max Index, Table is full and no need to check for additional events

15

lbl_EventNext
LBL

Display Event Removed
EventRemoved
/

Event Table Index
ADD
Source A EVi 0
Source B 1
Dest EVi 0

Event Table Index
LE
Source A EVi 0
Source B CFG.EV.mi <EDC.CFG.EV.mi> 9

lbl_EventRemoveLoop
JMP

lbl_Exit
JMP

Add New Events after the Existing Events
When EventRemoveLoop ends, EVi will point to first open position in Event Table

16

lbl_EventAdd
LBL

NOP

Exit without adding Events if Equipment Display None

17

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

lbl_Exit
JMP

Update Equip Events Table for HMI Displays

Initialize Event Search Array and Controls
Compare Inhibit bit set to enable 'One mismatch at a time' mode

18

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp
RES

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.IN
L

Search EQ Callout for Display Events FBC (Compare)
CPT
Dest EventFbcCmp.LEN 2080
Expression CFG.CalloutARR.n*32

Search EQ Callout for Display Events FBC (Compare)
MOVE
Source 0
Dest EventFbcCmp.POS 2080

Search EQ Callout for Display Events FBC (Results) EventFbcRes
RES

Search EQ Callout for Display Events FBC (Results)
MOVE
Source CFG.EV.n <EDC.CFG.EV.n> 10
Dest EventFbcRes.LEN 10

Search EQ Callout for Display Events FBC (Results)
MOVE
Source 0
Dest EventFbcRes.POS 0

Event File Bit Comparer (FBC)
Search Bit Array for On Event

19

lbl_EventAddLoop
LBL

Search EQ Callout for Display Events FBC (Results)
FBC
Source TempCallout[0]
Reference ZeroArray[0]
Result EventResArray[0]
Cmp. Control EventFbcCmp
Length 2080
Position 2080
Result Control EventFbcRes
Length 10
Position 0
EN
DN
FD
IN
ER

Update Equip Events Table for HMI Displays

If Event Found, convert Bit Number to Element Number and check Event Array and determine if exising or new
Element Number = (Word * 100) + Bit

20

Search EQ Callout for Display Events FBC (Compare)
EventFbcCmp.FD

Search EQ Callout for Display Events FBC (Results)
LIMIT
Low Limit 1
Test EventFbcRes.POS 0
High Limit CFG.EV.n <EDC.CFG.EV.n> 10

Event FBC Result Array Position
SUB
Source A EventFbcRes.POS 0
Source B 1
Dest EventResPos 0

Event FBC Result
MOVE
Source EventResArray[EventResPos] 1061
Dest EventRes 1265

Event Word Index
CPT
Dest WDi 39
Expression TRUNC(EventRes/32)

Event Bit Index
CPT
Dest BITi 17
Expression EventRes MOD 32

Event Element
CPT
Dest Element 3917
Expression WDi*100+BITi

PreDefined Event No Display if not Fault, Alarm or Alert
(Event Color is added in HMI code to save memory)
v3.5 moved v2_0 inside branch; always honor NoDisp

21

Event Word Index
EQ
Source A WDi 39
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

PreDefined Event Mask DtType Fault Mask
EDC_EventMaskPreDef.Fault.[BITi]

PreDefined Event Mask DtType Alarm Mask
EDC_EventMaskPreDef.Alarm.[BITi]

PreDefined Event Mask DtType Alert Mask
EDC_EventMaskPreDef.Alert.[BITi]

PreDefined Event Mask No Display Option
EDC_EventMaskPreDef.NoDisp.[BITi]

PreDefined No-Display Event
EventNoDisplay_PreDef

Update Equip Events Table for HMI Displays

EqDefined Event No Display if not Fault, Alarm or Alert
(Event Color is added in HMI code to save memory)
v3.5 moved v2_0 inside branch; always honor NoDisp

22

Event Word Index
GT
Source A WDi 39
Source B 0

Equip-Defined Event Word Index
SUB
Source A WDi 39
Source B 1
Dest EqDef_WDi 38

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Equipment Event Fault Mask
EquipMaskFault[EqDef_WDi].[BITi]

Equipment Event Alarm Mask
EquipMaskAlarm[EqDef_WDi].[BITi]

Equipment Event Alert Mask
EquipMaskAlert[EqDef_WDi].[BITi]

Equipment Event No Display Mask
EquipMaskNoDisp[EqDef_WDi].[BITi]

EqDefined No-Display Event
EventNoDisplay_EqDef

Update Equip Events Table for HMI Displays

If Event Found, convert Bit Number to Element Number and check Event Array and determine if exising or new
Element Number = (Word * 100) + Bit

PreDefined Word 0, Bits 0-7 and Bit 17 are not Displayed Events
(OverallStart, ConvRunning, FixedStopPosition, FixedStopReq, TempStopReq, ProdStopReq, Fault, Alarm, Alert)
DHF029 - Added Alert to not Displayed Events

23

Search EQ Callout for Display Events FBC (Compare)
EventFbcCmp.FD

Search EQ Callout for Display Events FBC (Results)
LIMIT
Low Limit 1
Test EventFbcRes.POS 0
High Limit CFG.EV.n <EDC.CFG.EV.n> 10

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

PreDefined No-Display Event
EventNoDisplay_PreDef

EqDefined No-Display Event
EventNoDisplay_EqDef

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Event Word Index
NE
Source A WDi 39
Source B 0

Event Bit Index
NE
Source A BITi 17
Source B 14

Event Word Index
EQ
Source A WDi 39
Source B 0

Event Bit Index
LIMIT
Low Limit 0
Test BITi 17
High Limit 7

Event Bit Index
EQ
Source A BITi 17
Source B 17

Block 'No Display Event' from Event Viewers
EventNoDisplay

Update Equip Events Table for HMI Displays

PreDef 19 & 20 - Health Events from Server
Use Gateway Timestamp

24

Event Word Index

EQ	
Source A	WDi 39
Source B	0

Event Bit Index

LIMIT	
Low Limit	19
Test	BITi 17
High Limit	20

Health Event from Server
EventHealth

Update Equip Events Table for HMI Displays

If Event Found and within Display Range, search existing Event Display for duplicate
if not found, add Event to Display
if found - existing Event, resume search without adding

25

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.FD

Block 'No Display Event' from Event Viewers EventNoDisplay

Search Event Display for existing (already displayed, don't need to add) EventExistFSC (RES)

Event Table Index

GT	
Source A	EVi
	0
Source B	0

Search Event Display for existing (already displayed, don't need to add)

MOVE	
Source	EVi
	0
Dest	EventExistFSC.LEN
	1

Search Event Display for existing (already displayed, don't need to add)

MOVE	
Source	0
Dest	EventExistFSC.POS
	0

Search Event Display for existing (already displayed, don't need to add)

FSC		
Control	EventExistFSC	EN
Length	1	DN
Position	0	ER
Mode	ALL	
Expression	Element=Equip.Event[EventExistFSC.POS].Element	

Search Event Display for existing (already displayed, don't need to add) EventExistFSC.FD

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element (Word*100) + Bit

MOVE	
Source	Element
	3917
Dest	Equip.Event[EVi].Element
	-1

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element Word

MOVE	
Source	WDi
	39
Dest	Equip.Event[EVi].Word
	-1

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element Bit

MOVE	
Source	BITi
	17
Dest	Equip.Event[EVi].Bit
	-1

Health Event from Server EventHealth

Local Copy of Equip Array Tag for LNi, EQi Transaction Time

COP	
Source	CalloutMsg.TransTime
Dest	Equip.Event[EVi].TransTime
Length	1

Event Table Index

ADD	
Source A	EVi
	0
Source B	1
Dest	EVi
	0

Health Event from Server EventHealth

Local Copy of Equip Array Tag for LNi, EQi Transaction Time

COP	
Source	CLX.TransTime
	<EDC.CLX.TransTime>
Dest	Equip.Event[EVi].TransTime
Length	1

Update Equip Events Table for HMI Displays

If Event Found and not within Display Range, decrement Result Array to make room for a different result
(otherwise Result Array fills up with Events that aren't displayed)

26

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.FD

Block 'No Display Event' from Event Viewers EventNoDisplay

Search EQ Callout for Display Events FBC (Results)

SUB
Source A EventFbcRes.POS[0]
Source B 1
Dest EventFbcRes.POS[0]

Search EQ Callout for Display Events FBC (Results)

LT
Source A EventFbcRes.POS[0]
Source B 0

Search EQ Callout for Display Events FBC (Results)

MOVE
Source 0
Dest EventFbcRes.POS[0]

If FBC Compare and Result Array not Done, AND Event Table not full -- resume the FBC Compare to look for another Event

27

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.DN

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.ER

Search EQ Callout for Display Events FBC (Results) EventFbcRes.DN

Search EQ Callout for Display Events FBC (Results) EventFbcRes.ER

Event Table Index

LE
Source A EVi[0]
Source B CFG.EV.mi <EDC.CFG.EV.mi> 9

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.EN

U

lbl_EventAddLoop

JMP

Exit Routine

28

lbl_Exit

LBL

NOP

(End)

Calculate Equipment Data and Operational Availability (OA)

Calculate Equipment Operational Availability (OA)

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

v1.6 Separate Line and Equipment Downtime

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 140
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Faster to JMP over AOI if not scanned
(AOI uses scan time even when Rung is scanned false)

2

Reset Summary Data for this Line
SumDataReset
/

lbl_SkipReset
JMP

Shift Reset for Equipment OA
Fault & Alarm Time are not reset at Shift Change

3

Reset Summary Data for this Line
SumDataReset

External Data Job Count

CLR
Dest EDC_CycleCount[LNi,EQi] ??

Data Reset for Line, Group, Equip

aoiEDC_DataReset
aoiEDC_DataReset aoiDataReset ...
Data Equip.Data
DataCtlPrev Equip.DataCTLprev

Exit if EQID out of range

4

lbl_SkipReset
LBL

Calulated EQID Index within array limits
EQID_IndexOK
/

lbl_Exit
JMP

Calculate Equipment Data and Operational Availability (OA)

If Offline, load Group & Line Indexes
Offline Equipment can be configured to use any Line for OverallStart/Running signals
Offline Equipment can be configured to be in any Group for HMI Display
DN006 - support multiple Offline Lines

5

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Offline Equipment Group Index
MOVE
Source Equip.CFGoffline.Group 0
Dest EquipGRPi 2

Offline Equipment Group Index
LT
Source A EquipGRPi 2
Source B 0

Offline Equipment Group Index
GT
Source A EquipGRPi 2
Source B CFG.GRP.ei <EDC.CFG.GRP.ei> 3

Offline Equipment Group Index
CLR
Dest EquipGRPi 2

Copy EDC_EQCallout Predefined to a local tag for named references
DN006 - support multiple Offline Lines

GroupDataSTS - accumulates overlapping Status of Downtime types (Tool Change, Options, Full, Short, ProdDelay)
GroupDataCTL - contains exclusive Control of Downtime Type (generated after all Group-Equipment processed)
STS & CTL are needed to handle seal-in of existing Downtime Type

6

Local copy of Callout Table Predefined for EQID
COP
Source EDC_EqCallout[LNi,EQi].PreDef
Dest PreDef
Length 1

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi
COP
Source EDC.Offline.DataSTS[EquipGRPi]
Dest GrpDataSTS
Length 1

Local Copy of Group Array Tag for GRPi
COP
Source EDC.Offline.DataCTL[EquipGRPi]
Dest GrpDataCTL
Length 1

Calculate Equipment Data and Operational Availability (OA)

External Data Trigger
Cycle Counts

used by Telit to trigger Robot data reads
and for Summary Data

7

Local copy of Callout Table Predefined for EQID Overload is ON
PreDef._12

Local Copy of Equip Array Tag for LNi, EQi Cycle Count External Data 1-Shot
Equip.DataCTL.CycleCountONS
ONS

External Data Job Count

ADD
Source A EDC_CycleCount[LNi,EQi] ??
Source B 1
Dest EDC_CycleCount[LNi,EQi] ??

Calculate Equipment Data and Operational Availability (OA)

Equipment Data - EDT/LDT

Determine Worktime and Stop status
EDT - Equip Downtime - EQ Not Ready
LDT - Line Downtime - Line Not Running

Equipment Producing (OR%)
Equipment Available (OA%)

8

Local copy of Callout Table Predefined for EQID Line Production Work Time
PreDef._00

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local copy of Callout Table Predefined for EQID Temp Stop Request
PreDef._04

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OR
Equip.DataCTL.EdtOR

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA
Equip.DataCTL.EdtOA

Local copy of Callout Table Predefined for EQID EQ Fixed Stop Request
PreDef._03

Local copy of Callout Table Predefined for EQID Line Conveyor Stopped at Fixed Stop Position (L10)
PreDef._02

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Local copy of Callout Table Predefined for EQID Fault Status is ON (from EqDriver Fault Mask)
PreDef._06

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local copy of Callout Table Predefined for EQID Line Conveyor Running
PreDef._01

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local copy of Callout Table Predefined for EQID Temp Stop Request
PreDef._04

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OR
Equip.DataCTL.LdtOR

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA
Equip.DataCTL.LdtOA

Local copy of Callout Table Predefined for EQID

Local copy of Callout Table Predefined for EQID Line Conveyor

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0

Calculate Equipment Data and Operational Availability (OA)

EQ Fixed Stop Request PreDef._03

Stopped at Fixed Stop Position (L10) PreDef._02

(new PreDefined bits) Equip.CTL.CalloutMsgV2_0

Equipment Data - EDT Maint/Prod

Determine Maintenance/Production Stop status
EDT - Equip Downtime - EQ Not Ready

Equipment Producing (OR%)
Equipment Available (OA%)

Sep 2023 - seal in initial Maint/Prod for duration of Downtime

9

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OR Equip.DataCTL.EdtOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OR Equip.DataCTL.EdtProdOR

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed PreDef._14

Local copy of Callout Table Predefined for EQID Production Stop Request (from EqDriver Mask Prod/Maint Mask) PreDef._05

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR Equip.DataCTL.EdtMaintOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR Equip.DataCTL.EdtMaintOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA Equip.DataCTL.EdtOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OA Equip.DataCTL.EdtMaintOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR Equip.DataCTL.EdtMaintOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OR Equip.DataCTL.EdtProdOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA Equip.DataCTL.EdtOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OA Equip.DataCTL.EdtProdOA

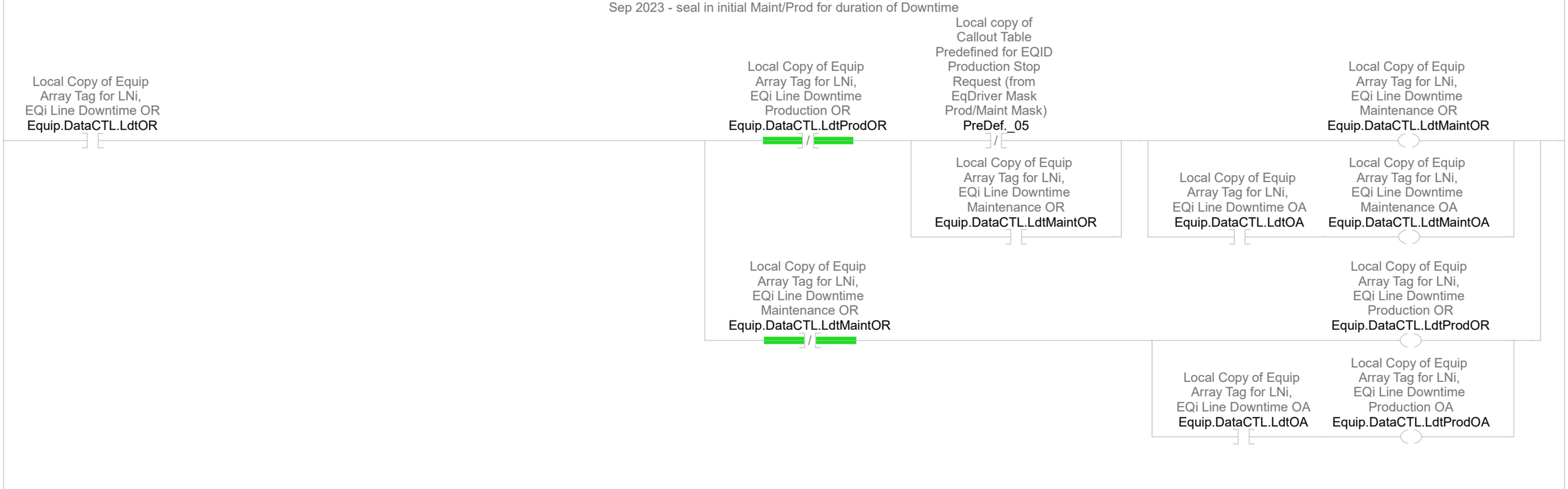
Calculate Equipment Data and Operational Availability (OA)

Equipment Data - LDT Maint/Prod

Determine Maintenance/Production Stop status
LDT - Line Downtime - Line Not Running

Equipment Producing (OR%)
Equipment Available (OA%)
Sep 2023 - seal in initial Maint/Prod for duration of Downtime

10

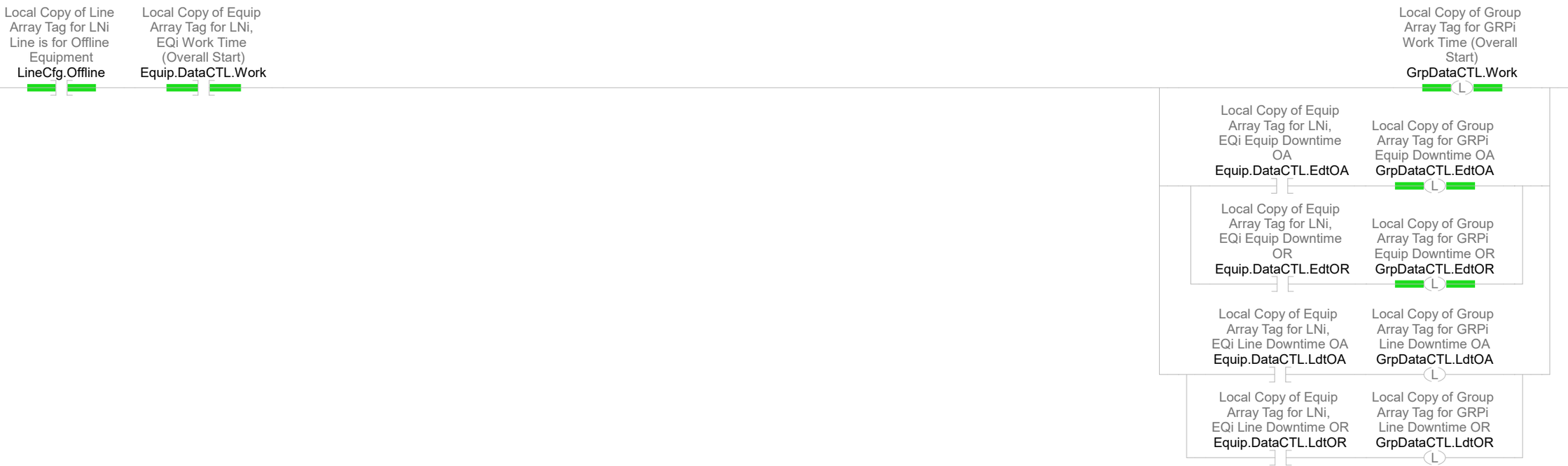


Calculate Equipment Data and Operational Availability (OA)

Group Data - EDT/LDT

Determine Worktime and Stop statusfor all EQID in Group
If Any Equip has Work/Stop, Group has Work/Stop (worst-case rollup)

11



Calculate Equipment Data and Operational Availability (OA)

Group Data - EDT Maint/Prod

Determine Maintenance/Production Downtime status for all EQID on Group
Feb 8, 2017 - changed Line Rollup to charge Maintenance if any EQ Maintenance

12

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OA
Equip.DataCTL.EdtMaintOA

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OA
GrpDataCTL.EdtMaintOA
L

Local Copy of Group Array Tag for GRPi Equip Downtime Production OA
GrpDataCTL.EdtProdOA
U

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR
Equip.DataCTL.EdtMaintOR

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OR
GrpDataCTL.EdtMaintOR
L

Local Copy of Group Array Tag for GRPi Equip Downtime Production OR
GrpDataCTL.EdtProdOR
U

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OA
Equip.DataCTL.EdtProdOA

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OA
GrpDataCTL.EdtMaintOA
/

Local Copy of Group Array Tag for GRPi Equip Downtime Production OA
GrpDataCTL.EdtProdOA
L

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OR
Equip.DataCTL.EdtProdOR

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OR
GrpDataCTL.EdtMaintOR
/

Local Copy of Group Array Tag for GRPi Equip Downtime Production OR
GrpDataCTL.EdtProdOR
L

Calculate Equipment Data and Operational Availability (OA)

Group Data - LDT Maint/Prod

Determine Maintenance/Production Stop status for all EQID on Group
Feb 8, 2017 - changed Line Rollup to charge Maintenance if any EQ Maintenance

13

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OA
Equip.DataCTL.LdtMaintOA

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OA
GrpDataCTL.LdtMaintOA

Local Copy of Group Array Tag for GRPi Line Downtime Production OA
GrpDataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OR
Equip.DataCTL.LdtMaintOR

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OR
GrpDataCTL.LdtMaintOR

Local Copy of Group Array Tag for GRPi Line Downtime Production OR
GrpDataCTL.LdtProdOR

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OA
Equip.DataCTL.LdtProdOA

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OA
GrpDataCTL.LdtMaintOA

Local Copy of Group Array Tag for GRPi Line Downtime Production OA
GrpDataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OR
Equip.DataCTL.LdtProdOR

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OR
GrpDataCTL.LdtMaintOR

Local Copy of Group Array Tag for GRPi Line Downtime Production OR
GrpDataCTL.LdtProdOR

Add Equipment Data to Line Data
Worktime and LineDT use Rollup of all Equiment or a Single EQi designated for Line Data

14

Equipment array Index

EQ
Source A EQi
12
Source B EDC_CfgGateway.LdtEQi
0

Gateway Config Download Line Downtime Rollup Enable
EDC_CfgGateway.LdtRollupEnable

Add Equip Data to Line Data
AddToLDT

Calculate Equipment Data and Operational Availability (OA)

This Line's Last Enabled Equip Loop scan
(update Line Data rollups)

15

Equipment array Index

EQ	
Source A	EQi 12
Source B	LineCfg.EQ.ENei 99

This Line's Last Equipment
LineCfg_LastEQi

Equipment Maintenance
Line Maintenance rollup of all Equipment

June 2025 -- add Line Rollup rules to Maint Mode selection
if rollup disabled, include Equip Bypass in Line Maint Mode

16

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local Copy of Equip Array Tag for LNi, EQi Maintenance Mode
Equip.DataCTL.MaintMode

Local copy of Callout Table Predefined for EQID Production Stop Request (from EqDriver Mask Prod/Maint Mask)
PreDef._05
/

Add Equip Data to Line Data
AddToLDT

Gateway Config Download Line Downtime Rollup Enable
EDC_CfgGateway.LdtRollupEnable
/

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local Copy of Line Array Tag for LNi Maintenance Mode
LineDataCTL.MaintMode
L

Local copy of Callout Table Predefined for EQID Production Stop Request (from EqDriver Mask Prod/Maint Mask)
PreDef._05
/

Calculate Equipment Data and Operational Availability (OA)

Line Data - EDT/LDT

Determine Line Worktime and LineDT statusfor all EQID on Line
Worktime and LineDT use Rollup of all Equiment or a Single EQi designated for Line Data
EquipDT uses Rollup of all Equipment

17

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start) Equip.DataCTL.Work

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Work Time (Overall Start) LineDataCTL.Work (L)

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA Equip.DataCTL.EdtOA

Local Copy of Line Array Tag for LNi Equip Downtime OA LineDataCTL.EdtOA (L)

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OR Equip.DataCTL.EdtOR

Local Copy of Line Array Tag for LNi Equip Downtime OR LineDataCTL.EdtOR (L)

Add Equip Data to Line Data AddToLDT

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA Equip.DataCTL.LdtOA

Local Copy of Line Array Tag for LNi Line Downtime OA LineDataCTL.LdtOA (L)

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OR Equip.DataCTL.LdtOR

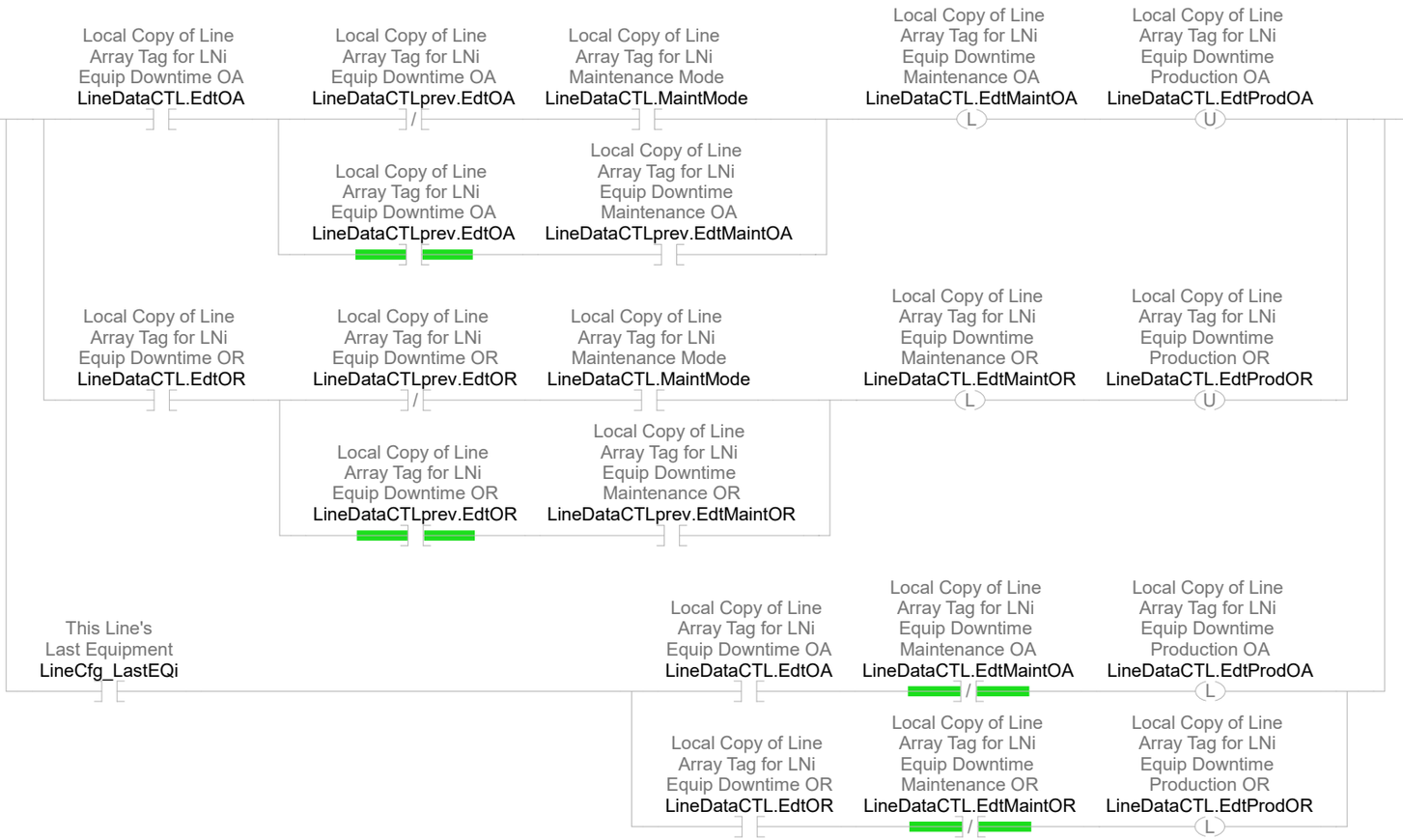
Local Copy of Line Array Tag for LNi Line Downtime OR LineDataCTL.LdtOR (L)

Calculate Equipment Data and Operational Availability (OA)

Line Data - EDT Maint/Prod

Determine Line Maintenance/Production Downtime status for all EQID on Line
Seal-in Maint status at start of Downtime if any Equip has Maint status - otherwise charge Production

18

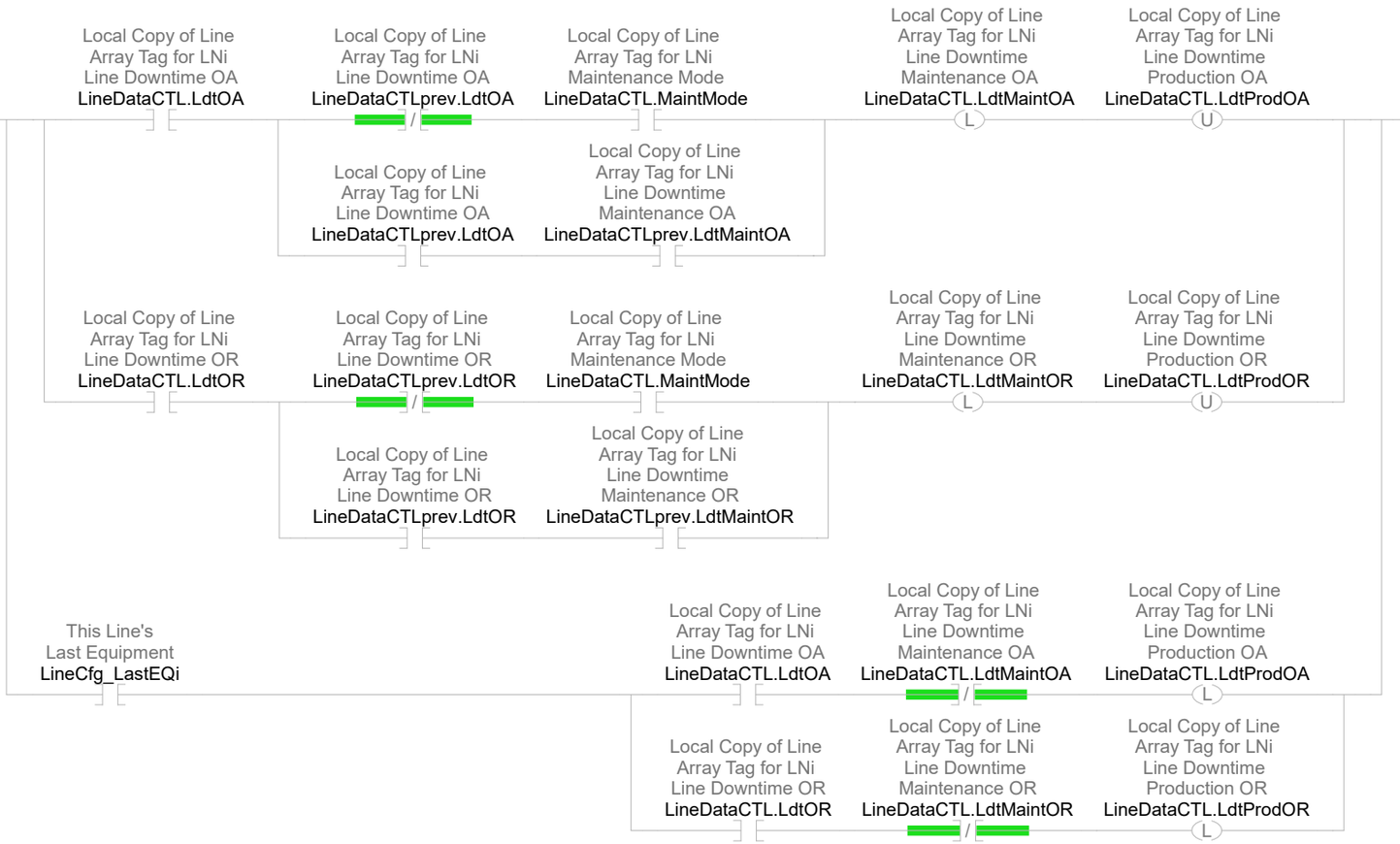


Calculate Equipment Data and Operational Availability (OA)

Line Data - LDT Maint/Prod

Determine Line Maintenance/Production Downtime status using Rollup of all Equiment or a Single EQi designated for Line Data
Seal-in Maint status at start of Downtime if any Equip has Maint status - otherwise charge Production

19



Calculate Equipment Data and Operational Availability (OA)

Determine Interlock status (Orange)
DHF029 Add Alert status
DN006 - support multiple Offline Lines

20

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Interlock Off Condition (Orange)
GrpDataCTL.InterlockOff
L

Local Copy of Group Array Tag for GRPi Fault Condition (Red)
GrpDataCTL.Fault
U

Local Copy of Group Array Tag for GRPi Alarm Condition (Yellow)
GrpDataCTL.Alarm
U

Local Copy of Group Array Tag for GRPi Alert Condition (Lt Blue)
GrpDataCTL.Alert
U

Add Equip Data to Line Data
AddToLDT

Local Copy of Line Array Tag for LNi Interlock Off Condition (Orange)
LineDataCTL.InterlockOff
L

Local Copy of Line Array Tag for LNi Fault Condition (Red)
LineDataCTL.Fault
U

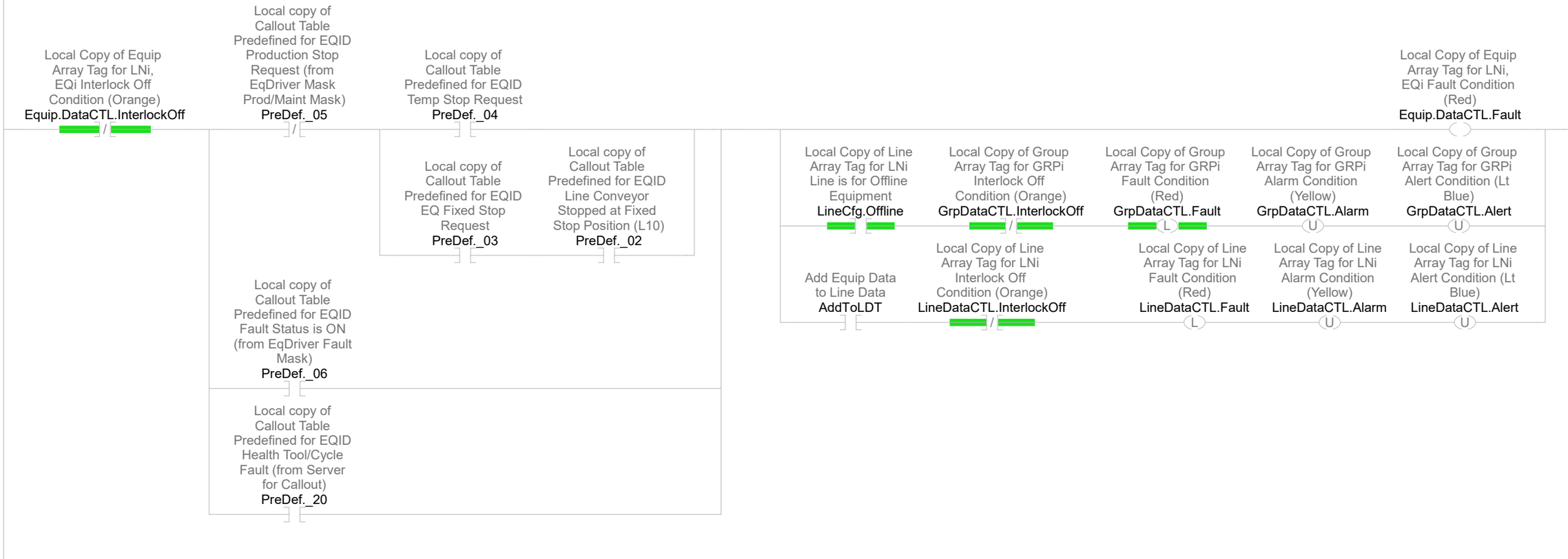
Local Copy of Line Array Tag for LNi Alarm Condition (Yellow)
LineDataCTL.Alarm
U

Local Copy of Line Array Tag for LNi Alert Condition (Lt Blue)
LineDataCTL.Alert
U

Calculate Equipment Data and Operational Availability (OA)

Determine Fault status (Red)
Offline if LNi = 0; Online if LNi <>0
DHF029 Add Alert status
DN006 - support multiple Offline Lines
DN014 - add EQ Tool Health Fault (PreDef.Spare20)

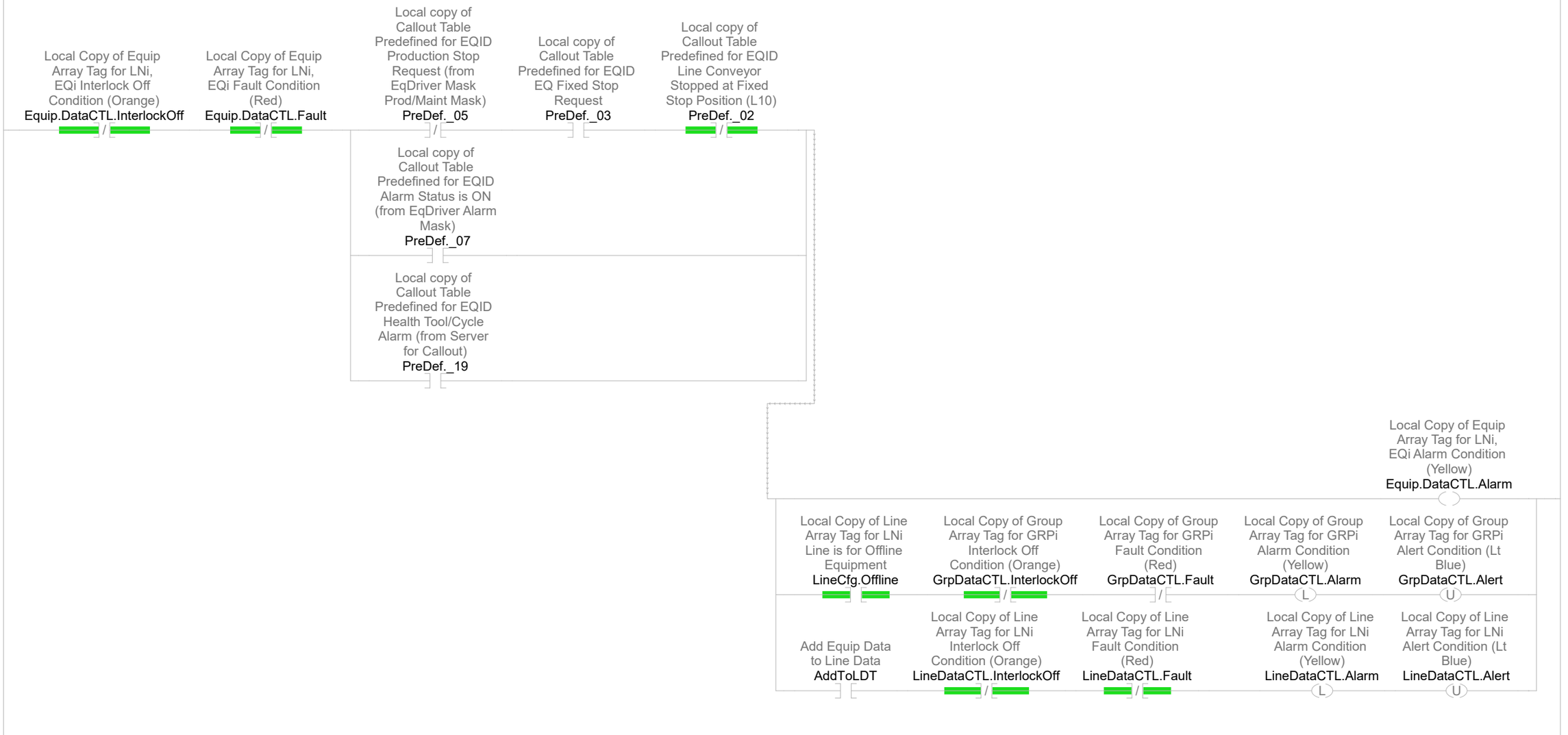
21



Calculate Equipment Data and Operational Availability (OA)

Determine Alarm status (Yellow)
DHF029 Add Alert status
DN006 - support multiple Offline Lines
DN014 - add EQ Tool Health Alarm (PreDef.Spare19)

22



Calculate Equipment Data and Operational Availability (OA)

DHF029 Determine Alert status (Lt Blue)
DN006 - support multiple Offline Lines

23

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange) Equip.DataCTL.InterlockOff

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red) Equip.DataCTL.Fault

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow) Equip.DataCTL.Alarm

Local copy of Callout Table Predefined for EQID Alert Status is ON (from EqDriver Alert Mask) PreDef._17

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue) Equip.DataCTL.Alert

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi Interlock Off Condition (Orange) GrpDataCTL.InterlockOff

Local Copy of Group Array Tag for GRPi Fault Condition (Red) GrpDataCTL.Fault

Local Copy of Group Array Tag for GRPi Alarm Condition (Yellow) GrpDataCTL.Alarm

Local Copy of Group Array Tag for GRPi Alert Condition (Lt Blue) GrpDataCTL.Alert

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Interlock Off Condition (Orange) LineDataCTL.InterlockOff

Local Copy of Line Array Tag for LNi Fault Condition (Red) LineDataCTL.Fault

Local Copy of Line Array Tag for LNi Alarm Condition (Yellow) LineDataCTL.Alarm

Local Copy of Line Array Tag for LNi Alert Condition (Lt Blue) LineDataCTL.Alert

Cycle

24

Local copy of Callout Table Predefined for EQID Overload is ON PreDef._12

Local Copy of Equip Array Tag for LNi, EQi Cycle Equip.DataCTL.Cycle

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data 1-shot Equip.DataCTL.CycleSendONS

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data Equip.DataCTL.CycleSend

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi Cycle GrpDataCTL.Cycle

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data GrpDataCTL.CycleSend

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Cycle LineDataCTL.Cycle

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data LineDataCTL.CycleSend

Calculate Equipment Data and Operational Availability (OA)

Assign Downtime Status if Worktime and Line Stopped w/ Request

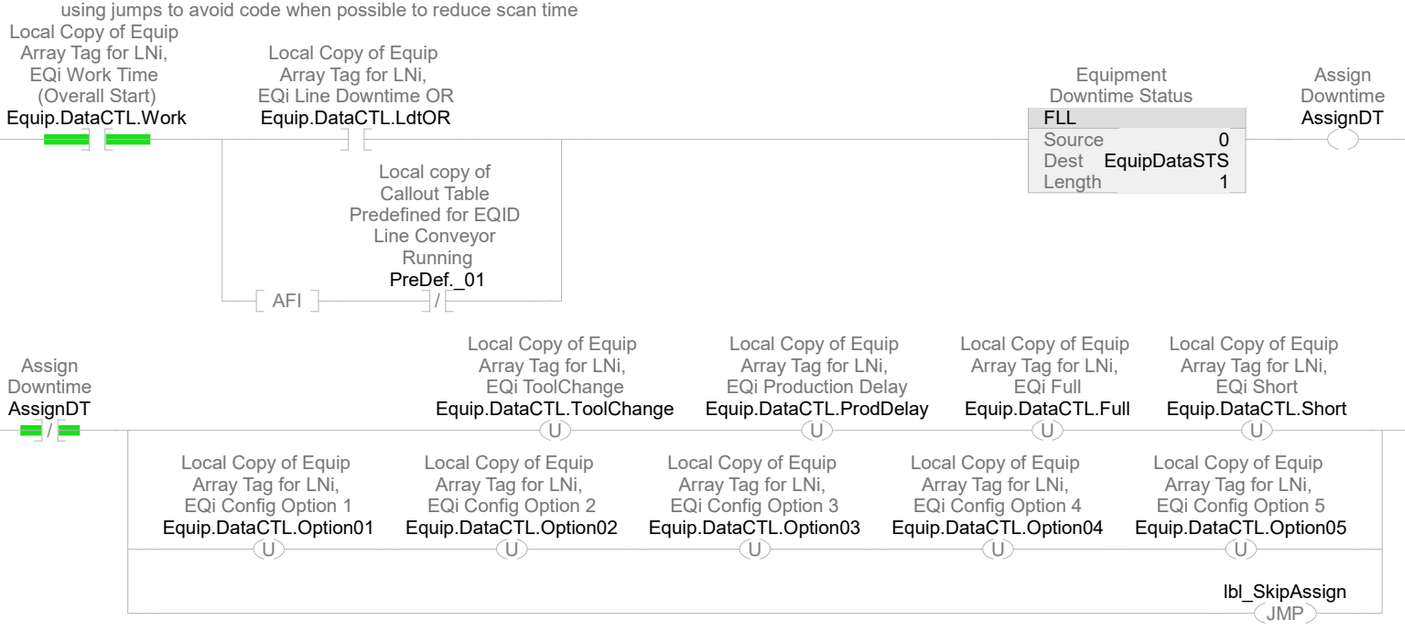
collect STS for Equip, Group and Line - more than 1 signal can be on in STS
the STS will be used to update CTL where only one signal can be on and existing DT items must be sealed-in while on

if Equip Running, no need to check for Downtime Assignment
clear the CTL bits and jump over the assignment code

Line must accumulate STS across multiple Equip
Group must accumulate STS across multiple Line/Equip

this code executes once for every configured Equip on every scan
using jumps to avoid code when possible to reduce scan time

25



Calculate Equipment Data and Operational Availability (OA)

Tool Change (ADC)
HIghest Priority - overrides all other DT

if Equip in Tool Change, no need to assign any other STS for this Equip, Group, Line

26

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA) PreDef._27

Local copy of Callout Table Predefined for EQID Tool Change (ADC) PreDef._28

Equipment Downtime Status ToolChange EquipDataSTS.ToolChange (L)

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi ToolChange GrpDataSTS.ToolChange (L)

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi ToolChange LineDataSTS.ToolChange (L)

lbl_AssignStsDone (JMP)

Production Delay

27

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA) PreDef._27

Local copy of Callout Table Predefined for EQID Delay (Prodution Hold, SubShort) PreDef._24

Equipment Downtime Status Production Delay EquipDataSTS.ProdDelay (L)

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi Production Delay GrpDataSTS.ProdDelay (L)

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Production Delay LineDataSTS.ProdDelay (L)

Calculate Equipment Data and Operational Availability (OA)

Full

28

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local copy of Callout Table Predefined for EQID Full
PreDef._26

Equipment Downtime Status Full
EquipDataSTS.Full
L

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Full
GrpDataSTS.Full
L

Add Equip Data to Line Data
AddToLDT

Local Copy of Line Array Tag for LNi Full
LineDataSTS.Full
L

Short

29

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local copy of Callout Table Predefined for EQID Short
PreDef._25

Equipment Downtime Status Short
EquipDataSTS.Short
L

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Short
GrpDataSTS.Short
L

Add Equip Data to Line Data
AddToLDT

Local Copy of Line Array Tag for LNi Short
LineDataSTS.Short
L

Skip Option Check if EQID Event Lookup Index is not good
Skip Option Check if not InDemand

30

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA
Equip.DataCTL.LdtOA
/

lbl_SkipOptions
JMP

Calculate Equipment Data and Operational Availability (OA)

Option 1
Rollup to Line from all Equip (ignoare AddToLDT)

31

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.Status

Equipment
Downtime Status
Config Option 1
EquipDataSTS.Option01
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 1
GrpDataSTS.Option01
L

Local Copy of Line
Array Tag for LNi
Config Option 1
LineDataSTS.Option01
L

Option 2
Rollup to Line from all Equip (ignoare AddToLDT)

32

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.Status

Equipment
Downtime Status
Config Option 2
EquipDataSTS.Option02
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 2
GrpDataSTS.Option02
L

Local Copy of Line
Array Tag for LNi
Config Option 2
LineDataSTS.Option02
L

Calculate Equipment Data and Operational Availability (OA)

Option 3
Rollup to Line from all Equip (ignoare AddToLDT)

33

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.Status

Equipment
Downtime Status
Config Option 3
EquipDataSTS.Option03
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 3
GrpDataSTS.Option03
L

Local Copy of Line
Array Tag for LNi
Config Option 3
LineDataSTS.Option03
L

Option 4
Rollup to Line from all Equip (ignoare AddToLDT)

34

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.Status

Equipment
Downtime Status
Config Option 4
EquipDataSTS.Option04
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 4
GrpDataSTS.Option04
L

Local Copy of Line
Array Tag for LNi
Config Option 4
LineDataSTS.Option04
L

Calculate Equipment Data and Operational Availability (OA)

Option 5
Rollup to Line from all Equip (ignoare AddToLDT)

35

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt05.Status

Equipment
Downtime Status
Config Option 5
EquipDataSTS.Option05
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 5
GrpDataSTS.Option05
L

Local Copy of Line
Array Tag for LNi
Config Option 5
LineDataSTS.Option05
L

Skip Option Check if EQID Event Lookup Index is not good

36

lbl_SkipOptions
LBL

NOP

Downtime Assign is Done
get Equip Priority Downtime (CTL) from Status (STS)

37

lbl_AssignStsDone
LBL

Prioritize STS
for CTL
aoiEDC_DataCtl
aoiEDC_DataCtl aoiDataCtl ...
DataSts EquipDataSTS
DataCtl Equip.DataCTL

skip Assign if Equip Running

38

lbl_SkipAssign
LBL

NOP

Calculate Equipment Data and Operational Availability (OA)

Equip Data Increment and OA
Add Incremental Time to Worktime, Downtime that are ON
Add Incremental Time to to Current Alarm or Fault Duration
Calculate OA
DN006 - support multiple Offline Lines

39

Increment Data

aoiEDC_DataInc
aoiEDC_DataInc aoiDataInc
Data Equip.Data
DataCtl Equip.DataCTL
DataCtlPrev Equip.DataCTLprev
Increment LineCtl.DataInc

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Equipment Data Collection Offline Group Data Status

COP
Source GrpDataSTS
Dest EDC.Offline.DataSTS[EquipGRPi]
Length 1

Equipment Data Collection Offline Group Data Control

COP
Source GrpDataCTL
Dest EDC.Offline.DataCTL[EquipGRPi]
Length 1

Exit Routine

lbl_Exit

40

LBL

NOP

(End)

Offline Group Data Initialize

Initialize, Reset or Update Offline Group Operational Availability (OA) Controls

One set of Interfaces is shared by all Group
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 141
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Initialize Group Data Control Word

2

Local Copy of Offline Array Tag for LNi

FLL
Source 0
Dest OfflineDataCTL
Length 1

Offline Group Data Initialize

v3.5 Preserve the Downtime Type seal-in

3

Equipment Data Collection Offline Group Data Control	Local Copy of Offline Array Tag for LNi
EDC.Offline.DataCTL[GRPi].ToolChange	OfflineDataCTL.ToolChange (L)
EDC.Offline.DataCTL[GRPi].Option01	OfflineDataCTL.Option01 (L)
EDC.Offline.DataCTL[GRPi].Option02	OfflineDataCTL.Option02 (L)
EDC.Offline.DataCTL[GRPi].Option03	OfflineDataCTL.Option03 (L)
EDC.Offline.DataCTL[GRPi].Option04	OfflineDataCTL.Option04 (L)
EDC.Offline.DataCTL[GRPi].Option05	OfflineDataCTL.Option05 (L)
EDC.Offline.DataCTL[GRPi].ProdDelay	OfflineDataCTL.ProdDelay (L)
EDC.Offline.DataCTL[GRPi].Full	OfflineDataCTL.Full (L)
EDC.Offline.DataCTL[GRPi].Short	OfflineDataCTL.Short (L)

Offline Group Data Initialize

Nov 24, 2025 - Preserve the CycleSend signal for sending Summary Data updates when still cycling after WorkTime is off

4

Equipment Data Collection Offline Group Data Control
EDC.Offline.DataCTL[GRPi].CycleSend

Local Copy of Offline Array Tag for LNi Cycle Count Send Summary Data
OfflineDataCTL.CycleSend
L

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

OfflineDataSTS is reloaded each scan by the Line/Equip Loop for comparison to OfflineDataCtlprev
OfflineDataCTL is a mixed bag -- some bits are cleared and some are retained for seal-in

5

Equipment Data Collection Offline Group Data Status

FLL	
Source	0
Dest	EDC.Offline.DataSTS[GRPi]
Length	1

Equipment Data Collection Offline Group Data Control

COP	
Source	OfflineDataCTL
Dest	EDC.Offline.DataCTL[GRPi]
Length	1

(End)

Offline Group Data Reset

Initialize, Reset or Update Offline Group Operational Availability (OA) Controls

One set of Interfaces is shared by all Group
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 142
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Copy Offline Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.Data[GRPi]
Dest OfflineData
Length 1

Shift Reset of Group Data

3

Data Reset for Line, Group, Equip

aoiEDC_DataReset
aoiEDC_DataReset aoiDataReset ...
Data OfflineData
DataCtlPrev OfflineDataCTLprev

Offline Group Data Reset

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

4

Equipment Data Collection Offline Group Data (1-Trim, 2-Chassis, 3-Final)

COP	
Source	OfflineData
Dest	EDC.Offline.Data[GRPi]
Length	1

(End)

Offline Group Data Update

Initialize, Reset or Update Offline Group Operational Availability (OA) Controls

One set of Interfaces is shared by all Group
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 143
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Copy Offline Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.DataSTS[GRPi]
Dest OfflineDataSTS
Length 1

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.DataCTL[GRPi]
Dest OfflineDataCTL
Length 1

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.DataCTLprev[GRPi]
Dest OfflineDataCTLprev
Length 1

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.Data[GRPi]
Dest OfflineData
Length 1

Offline Group Data Increment and OA
Add Incremental Time to Worktime, Downtime that are ON
Add Incremental Time to to Current Alarm or Fault Duration
Calculate OA

3

Prioritize STS for CTL

aoiEDC_DataCtl
aoiEDC_DataCtl aoiDataCtl (...)
DataSts OfflineDataSTS
DataCtl OfflineDataCTL

Increment Data

aoiEDC_DataInc
aoiEDC_DataInc aoiDataInc (...)
Data OfflineData
DataCtl OfflineDataCTL
DataCtlPrev OfflineDataCTLprev
Increment LineCtl.DataInc

Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OA% - InDemand Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

4

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work
/

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data
GrpDataCTL.CycleSend

Schedule New Hour
ScheduleNewHour

Local Copy of Offline Array Tag for LNi Work Time Ended
OfflineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 3
Dest SumMsg.Header.DataType 13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source GRPi 4
Dest SumMsg.Header.ID 3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source OfflineData.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source OfflineData.EdtOA.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source OfflineData.EdtOA.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source OfflineData.LdtOA.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source OfflineData.LdtOA.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP
Source OfflineData.Full
Dest SumMsg.Data.Full
Length 1

Equipment Data Summary Message 7 - Short

COP
Source OfflineData.Short
Dest SumMsg.Data.Short
Length 1

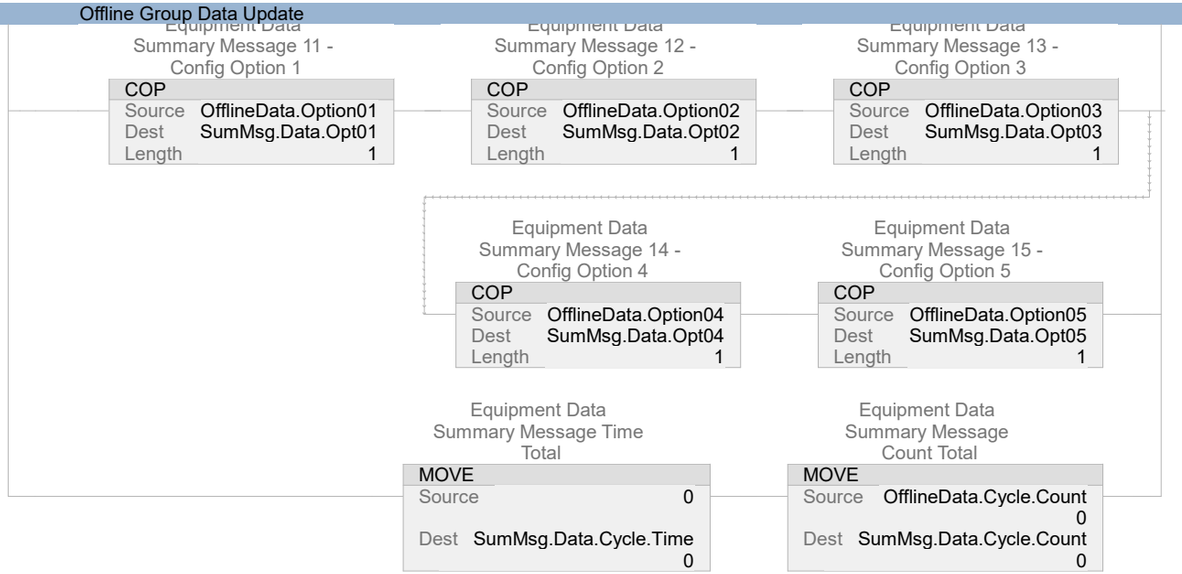
Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP
Source OfflineData.ProdDelay
Dest SumMsg.Data.ProdDelay
Length 1

Equipment Data Summary Message 10 - Tool Change (ADC)

COP
Source OfflineData.ToolChange
Dest SumMsg.Data.ToolChange
Length 1

Offline Group Data Update



Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Group Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OR% - All Downtime

5

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

RingBuffer Modulo Result

MOD
Source A GRPi
4
Source B 2
Dest BufferID
1

RingBuffer Modulo Result

EQ
Source A BufferID
1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OR% - All Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

6

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data
GrpDataCTL.CycleSend

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data
GrpDataCTL.CycleSend
U

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work
/

Schedule New Hour
ScheduleNewHour

Local Copy of Offline Array Tag for LNi Work Time Ended
OfflineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE	
Source	13
Dest	SumMsg.Header.DataType
	13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE	
Source	GRPi
	4
Dest	SumMsg.Header.ID
	3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP	
Source	OfflineData.Work
Dest	SumMsg.Data.Work
Length	1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP	
Source	OfflineData.EdtOR.Maint
Dest	SumMsg.Data.EdtMaint
Length	1

Equipment Data Summary Message 3 - Equip Downtime Production

COP	
Source	OfflineData.EdtOR.Prod
Dest	SumMsg.Data.EdtProd
Length	1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP	
Source	OfflineData.LdtOR.Maint
Dest	SumMsg.Data.LdtMaint
Length	1

Equipment Data Summary Message 5 - Line Downtime Production

COP	
Source	OfflineData.LdtOR.Prod
Dest	SumMsg.Data.LdtProd
Length	1

Equipment Data Summary Message 6 -Full

COP	
Source	OfflineData.Full

Equipment Data Summary Message 7 - Short

COP	
Source	OfflineData.Short

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP	
Source	OfflineData.ProdDelay

Equipment Data Summary Message 10 - Tool Change (ADC)

COP	
Source	OfflineData.ToolChange

Offline Group Data Update

		Dest SumMsg.Data.Full Length 1	Dest SumMsg.Data.Short Length 1	Dest SumMsg.Data.ProdDelay Length 1	Dest SumMsg.Data.ToolChange Length 1
Equipment Data Summary Message 11 - Config Option 1	Equipment Data Summary Message 12 - Config Option 2	Equipment Data Summary Message 13 - Config Option 3	Equipment Data Summary Message 14 - Config Option 4	Equipment Data Summary Message 15 - Config Option 5	
COP Source OfflineData.Option01 Dest SumMsg.Data.Opt01 Length 1	COP Source OfflineData.Option02 Dest SumMsg.Data.Opt02 Length 1	COP Source OfflineData.Option03 Dest SumMsg.Data.Opt03 Length 1	COP Source OfflineData.Option04 Dest SumMsg.Data.Opt04 Length 1	COP Source OfflineData.Option05 Dest SumMsg.Data.Opt05 Length 1	
			Equipment Data Summary Message Time Total	Equipment Data Summary Message Count Total	
			MOVE Source 0 Dest SumMsg.Data.Cycle.Time 0	MOVE Source OfflineData.Cycle.Count 0 Dest SumMsg.Data.Cycle.Count 0	

Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Group Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OA% - InDemand Downtime

7

SumDataLoadRB

Load Summary Data Message

JSR	
Routine Name	_150_SumDataMsg

RingBuffer Modulo Result

MOD	
Source A	GRPi 4
Source B	2
Dest	BufferID 1

RingBuffer Modulo Result

EQ	
Source A	BufferID 1
Source B	1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR	
Routine Name	_151_SumLoadRB01

UseSumRB01 (/)

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR	
Routine Name	_152_SumLoadRB02

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

8

Equipment Data Collection Offline Group Data Control

COP	
Source	OfflineDataCTL
Dest	EDC.Offline.DataCTL[GRPi]
Length	1

Equipment Data Collection Offline Group Data Control previous values

COP	
Source	OfflineDataCTLprev
Dest	EDC.Offline.DataCTLprev[GRPi]
Length	1

Equipment Data Collection Offline Group Data (1-Trim, 2-Chassis, 3-Final)

COP	
Source	OfflineData
Dest	EDC.Offline.Data[GRPi]
Length	1

(End)

Load Summary Data Message

Load Equipment, Line, Group Summary Data into Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 150
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Increment Transaction Number

2

ADD
Source A SumTransactionNumber 7244
Source B 1
Dest SumTransactionNumber 7244

LT
Source A SumTransactionNumber 7244
Source B 1

GT
Source A SumTransactionNumber 7244
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

MOVE
Source 1
Dest SumTransactionNumber 7244

Load Summary Data Message

Load Message Header, Timestamp and Footer Data
SumMsg Header.ID, Header.DataType and Data are loaded by calling routine
Header.Spare is used for Schedule Hour 1-10
for Schedule Hour, load the SchedulePrev.Hour to properly handle hour changes; accum data is for the previous minute; data for the current minute hasn't been collected yet
(i.e. final data packet for hour 0 is sent at the start of hour 1, and the Schedule Hour must be 0 for this packet)

*** SPECIAL CASE ***
for end of shift closeout record - clear the fractional seconds to avoid assigning record to the next shift
(SQL is using Transaction Timestamp to determine ShiftDate, ShiftID - Gateway must send a shift closeout record to land exactly on the SQL ShiftWindowEnd datetime)

3

Equipment Data Summary Message Header - Transaction Number (must match Footer)

MOVE
Source SumTransactionNumber 7244
Dest SumMsg.Header.TransNum 7244

Equipment Data Summary Message Header - Command Number

MOVE
Source 0
Dest SumMsg.Header.CmdNum 0

Equipment Data Summary Message Header - Schedule Hour 0-10

MOVE
Source SchedulePrev.Hour 9
Dest SumMsg.Header.Hour 9

Equipment Data Summary Message Transaction Time Stamp

COP
Source CLX.TransTime <EDC.CLX.TransTime>
Dest SumMsg.TransTime
Length 1

Schedule Shift Reset

ScheduleShiftReset

Equipment Data Summary Message Fractional Seconds (.ss)

CLR
Dest SumMsg.TransTime.FracSec 1

Equipment Data Summary Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest SumMsg.Footer.Spare 0

Equipment Data Summary Message Footer - Transaction Number (must match Footer)

MOVE
Source SumTransactionNumber 7244
Dest SumMsg.Footer.TransNum 7244

(End)

Load Equipment,Line, Group Summary Data into Ring Buffer

Load Equipment, Line, Group Summary Data Message into Ring Buffer

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 151
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RingBufferSum01.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RingBufferSum01.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RingBufferSum01.WritePtrErr <EDC.RB_Sum01.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest RingBufferSum01.WritePtrErrTag <EDC.RB_Sum01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest RingBufferSum01.WritePtrErrRec <EDC.RB_Sum01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2413
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RingBufferSum01.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RingBufferSum01.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87

EDC Ring Buffer Control Buffer Full - Block Writes

RingBufferSum01.FullBlockWrite <EDC.RB_Sum01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RingBufferSum01.WriteBlockErr <EDC.RB_Sum01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RingBufferSum01.WriteBlockErrCnt <EDC.RB_Sum01.WriteBlockErrCnt> 0
Source B 1
Dest RingBufferSum01.WriteBlockErrCnt <EDC.RB_Sum01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2414
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

lbl_Exit
JMP

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec01[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 2

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec02[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 3

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec03[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 4

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec04[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 5

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec05[RingBufferSum01.WriteRecPtr]
Length 1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.WriteTagPtr
	<EDC.RB_Sum01.WriteTagPtr>
	1
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.WriteRecPtr
	<EDC.RB_Sum01.WriteRecPtr>
	88
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.ReadTagPtr
	<EDC.RB_Sum01.ReadTagPtr>
	1
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.ReadRecPtr
	<EDC.RB_Sum01.ReadRecPtr>
	87
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2411
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	-1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 1
Dest RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RingBufferSum01.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Next Write Tag

ADD
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1
Dest RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RingBufferSum01.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

Load Equipment,Line, Group Summary Data into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RingBufferSum01.ShiftCnt <EDC.RB_Sum01.ShiftCnt> 113402
Source B 1
Dest RingBufferSum01.ShiftCnt <EDC.RB_Sum01.ShiftCnt> 113402

lbl_Exit

8 LBL

NOP

(End)

Load Equipment,Line, Group Summary Data into Ring Buffer

Load Equipment, Line, Group Summary Data Message into Ring Buffer

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 152
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B RingBufferSum02.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B RingBufferSum02.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RingBufferSum02.WritePtrErr <EDC.RB_Sum02.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest RingBufferSum02.WritePtrErrTag <EDC.RB_Sum02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest RingBufferSum02.WritePtrErrRec <EDC.RB_Sum02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2413
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B RingBufferSum02.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B RingBufferSum02.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45

EDC Ring Buffer Control Buffer Full - Block Writes
RingBufferSum02.FullBlockWrite <EDC.RB_Sum02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RingBufferSum02.WriteBlockErr <EDC.RB_Sum02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RingBufferSum02.WriteBlockErrCnt <EDC.RB_Sum02.WriteBlockErrCnt> 0
Source B 1
Dest RingBufferSum02.WriteBlockErrCnt <EDC.RB_Sum02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2414
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

lbl_Exit
JMP

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec01[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 2

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec02[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 3

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec03[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 4

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec04[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 5

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec05[RingBufferSum02.WriteRecPtr]
Length 1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.WriteRecPtr
	<EDC.RB_Sum02.WriteRecPtr>
	46
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.ReadTagPtr
	<EDC.RB_Sum02.ReadTagPtr>
	4
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.ReadRecPtr
	<EDC.RB_Sum02.ReadRecPtr>
	45
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2411
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	-1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B 1
Dest RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B RingBufferSum02.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Next Write Tag

ADD
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1
Dest RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B RingBufferSum02.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

Load Equipment,Line, Group Summary Data into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RingBufferSum02.ShiftCnt <EDC.RB_Sum02.ShiftCnt> 113130
Source B 1
Dest RingBufferSum02.ShiftCnt <EDC.RB_Sum02.ShiftCnt> 113130

lbl_Exit

8 LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR	
Input Par	Trans.Code
	<EDC.Trans.Code>
Input Par	Trans.Time
	<EDC.Trans.Time>
Input Par	Trans.LNi
	<EDC.Trans.LNi>
Input Par	Trans.EQi
	<EDC.Trans.EQi>
Input Par	Trans.Num
	<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	901
Dest	Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[901] -1502204510
Source B	1
Dest	Prog.RoutineScan[901] -1502204510

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

ADD
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 1
Dest: Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 0

Transaction Log Control -- Transaction Log Index

GE
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest: Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

ADD
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 1
Dest: Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A: Trans.LOGx <EDC.Trans.LOGx> 212
Source B: 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A: Trans.LOGx <EDC.Trans.LOGx> 212
Source B: CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest: Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

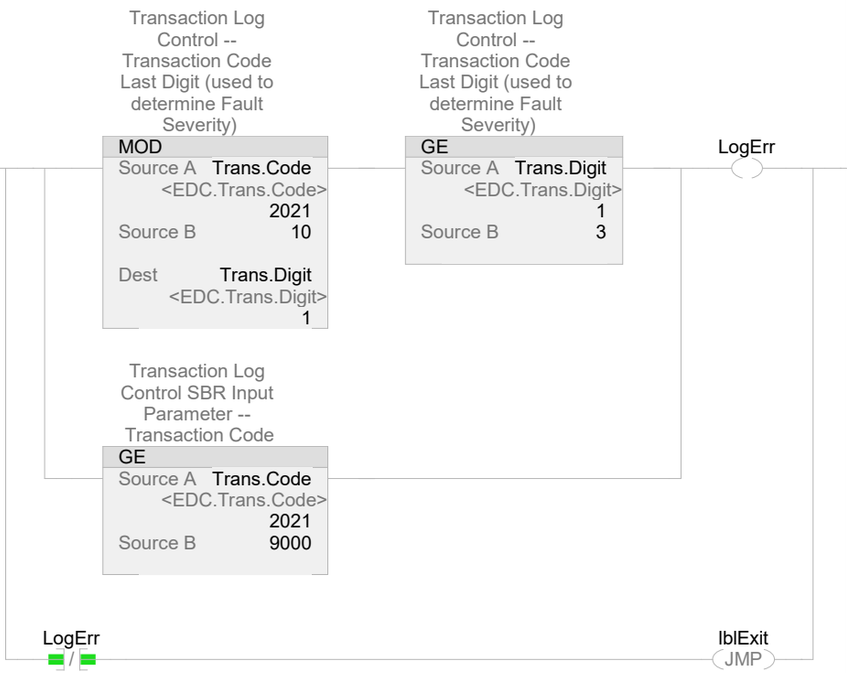
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

eAssy_EDC_Gway
MainTask
EquipRxLiveCallout
_101_Main
Ladder Diagram 1
_102_Fault
Ladder Diagram 8
_111_Config
Ladder Diagram 12
_120_LineLoop
Ladder Diagram 26
_130_EquipLoop
Ladder Diagram 39
_131_EquipCalloutRx
Ladder Diagram 54
_132_EquipCalloutRxLoad
Ladder Diagram 68
_133_EquipCalloutRxMapV1
Ladder Diagram 79
_134_EquipCalloutRxResponse
Ladder Diagram 80
_135_EquipEvents
Ladder Diagram 88
_140_EquipData
Ladder Diagram 103
_141_OfflineGroupDataInit
Ladder Diagram 127
_142_OfflineGroupDataReset
Ladder Diagram 130
_143_OfflineGroupData
Ladder Diagram 132
_150_SumDataMsg
Ladder Diagram 139
_151_SumLoadRB01
Ladder Diagram 141
_152_SumLoadRB02
Ladder Diagram 148
_901_TransLogSBR
Ladder Diagram 155